

Lecture 6

Work and Energy Principle

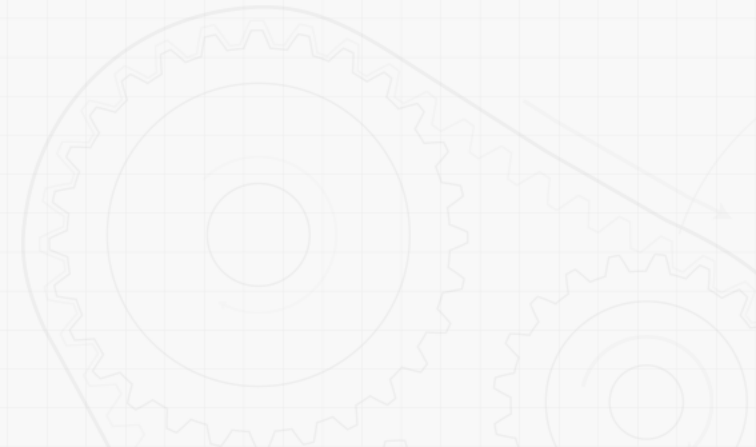
Prof. Pilwon Hur
MC2103
Fall 2026



6 Work and Energy Principle

Energy and Momentum Methods

The potential energy of the roller coaster car is converted into kinetic energy as it descends the track.



6 Work and Energy Principle

Energy and Momentum Methods

Impact tests are often analyzed by using momentum methods.



6 Work and Energy Principle

Introduction

Previously, problems dealing with the motion of particles were solved through the fundamental equation of motion,

$$\sum \vec{F} = m\vec{a}.$$

6 Work and Energy Principle

Introduction

⚙️ The current chapter introduces two additional methods of analysis.

**Method of
work and
energy**

➡️ Directly relates force, mass, velocity and displacement.

**Method of
impulse and
momentum**

➡️ Directly relates force, mass, velocity, and time.

6 Work and Energy Principle

Introduction

⚙ Approaches to Kinetics Problems

Forces and
Accelerations



Newton's Second
Law(last chapter)

$$\sum \vec{F} = m\vec{a}_G$$

Velocities and
Displacements



Work-Energy

$$T_1 + U_{1 \rightarrow 2} = T_2$$

Velocities and
Time

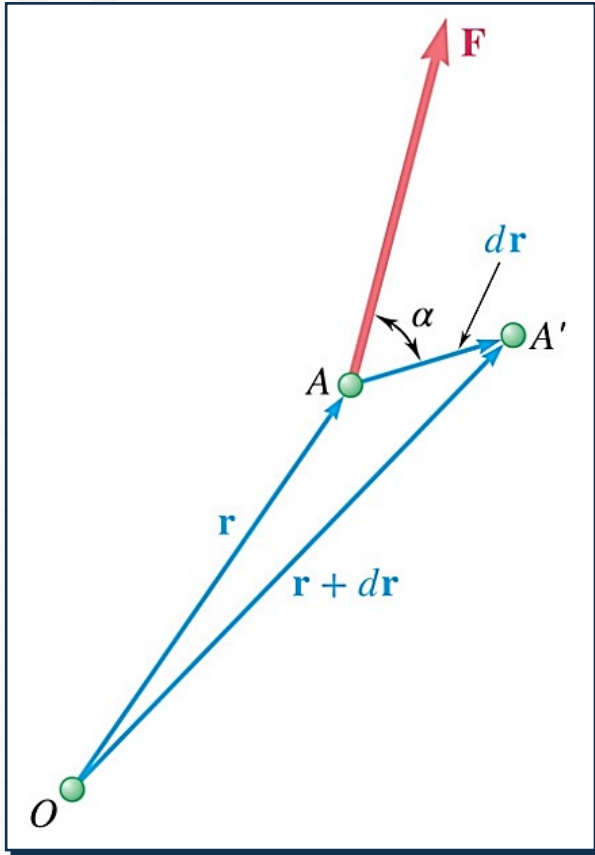


Impulse-
Momentum

$$m\bar{v}_1 + \int_{t_1}^{t_2} \vec{F} dt = m\bar{v}_2$$

6 Work and Energy Principle

Work of a Force



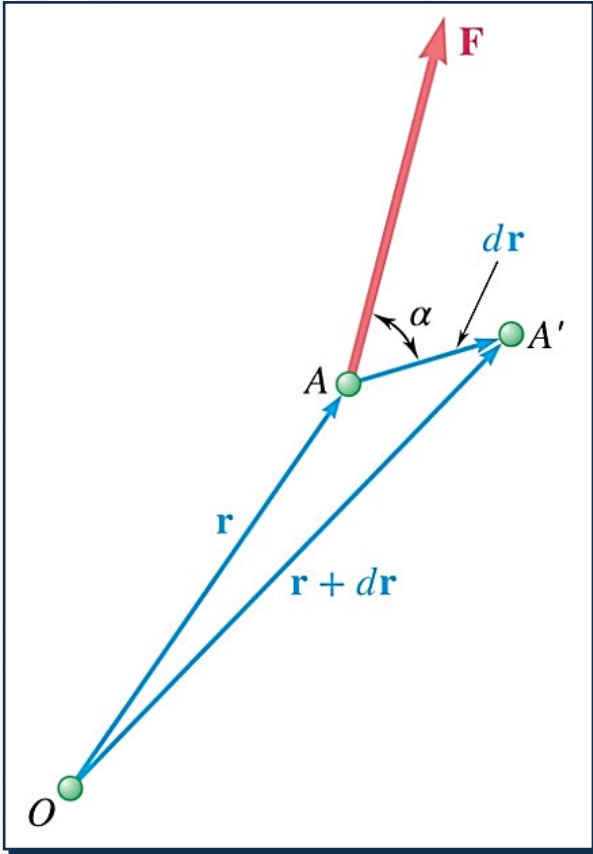
- Differential vector $d\vec{r}$ is the *particle displacement*.

Work of the force is

$$\begin{aligned} dU &= \vec{F} \cdot d\vec{r} \\ &= F ds \cos \alpha \\ &= F_x dx + F_y dy + F_z dz \end{aligned}$$

6 Work and Energy Principle

Work of a Force

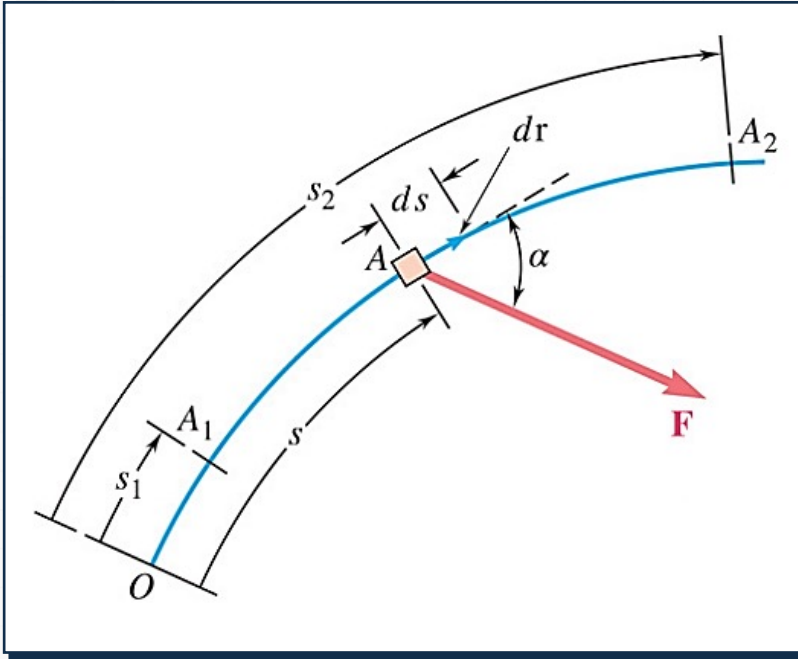


- Work is a *scalar* quantity.
 - It has magnitude and sign but not direction.
- Dimensions of work are length $\times$ force.
 - Units are

$$1 \text{ J(joule)} = (1\text{N})(1\text{m})$$

6 Work and Energy Principle

Work of a Force

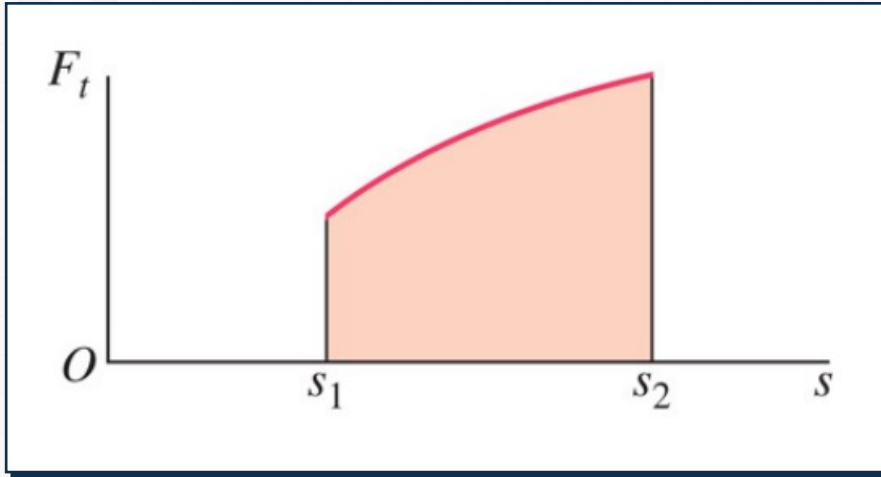


- Work of a force during a finite displacement,

$$\begin{aligned} U_{1 \rightarrow 2} &= \int_{A_1}^{A_2} \vec{F} \cdot d\vec{r} \\ &= \int_{s_1}^{s_2} (F \cos \alpha) ds = \int_{s_1}^{s_2} F_t ds \\ &= \int_{A_1}^{A_2} (F_x dx + F_y dy + F_z dz) \end{aligned}$$

6 Work and Energy Principle

Work of a Force

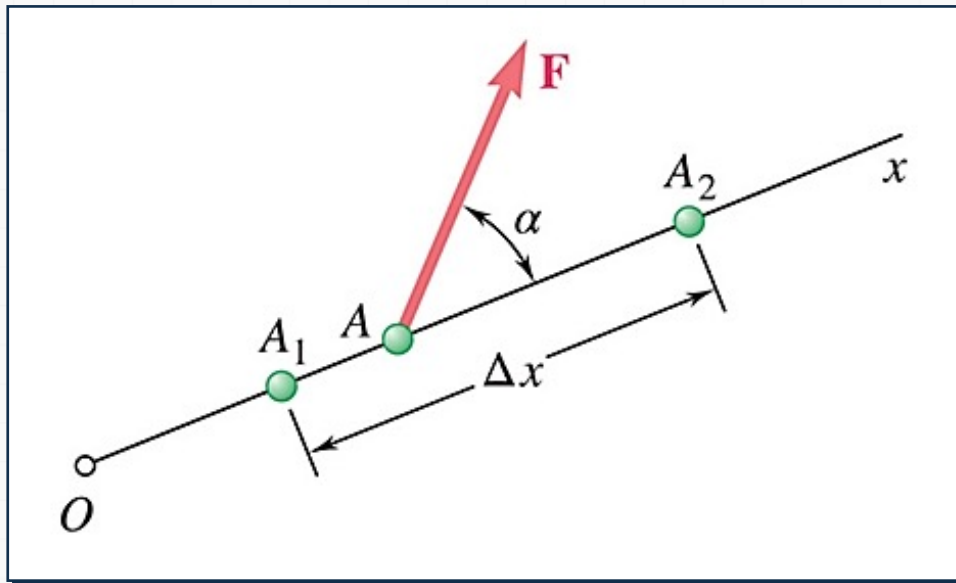


- Work is represented by the area under the curve of F_t plotted against s .
- F_t is the force in the direction of the displacement ds .

6 Work and Energy Principle

Work of a Force

⚙️ What is the work of a constant force in rectilinear motion?



a) $U_{1 \rightarrow 2} = F \Delta x$

b) $U_{1 \rightarrow 2} = (F \cos \alpha) \Delta x$

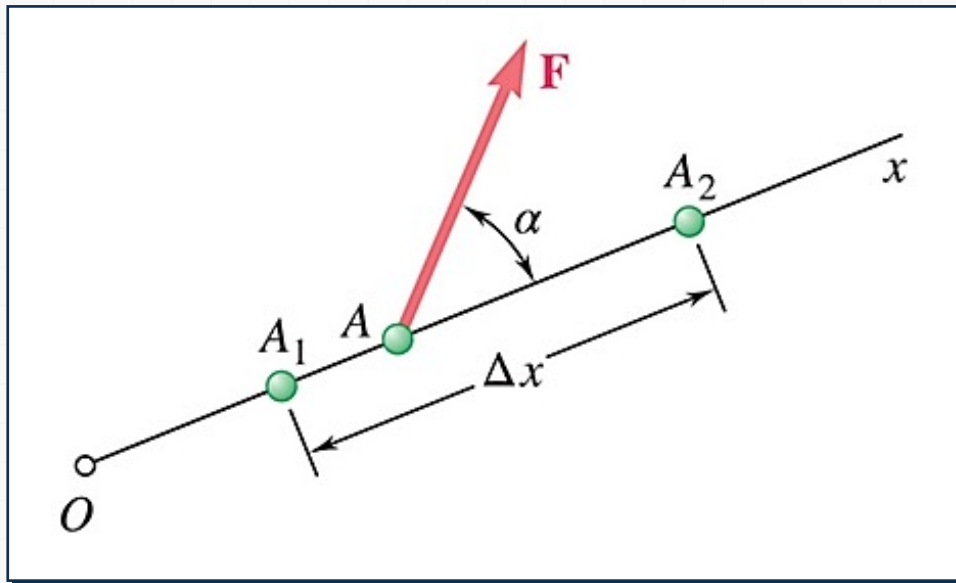
c) $U_{1 \rightarrow 2} = (F \sin \alpha) \Delta x$

d) $U_{1 \rightarrow 2} = 0$

6 Work and Energy Principle

Work of a Force

⚙️ What is the work of a constant force in rectilinear motion?



a) $U_{1 \rightarrow 2} = F \Delta x$

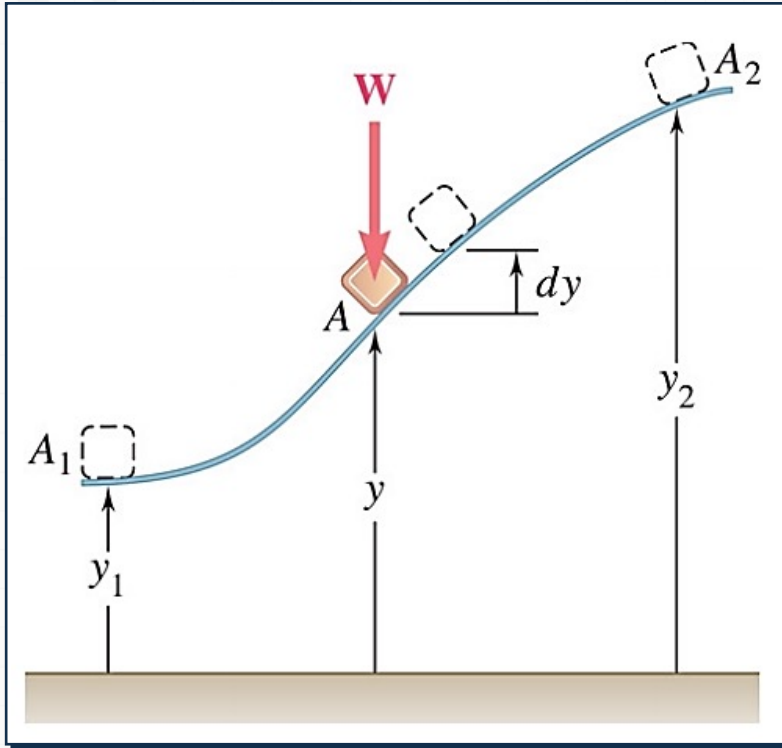
b) $U_{1 \rightarrow 2} = (F \cos \alpha) \Delta x$

c) $U_{1 \rightarrow 2} = (F \sin \alpha) \Delta x$

d) $U_{1 \rightarrow 2} = 0$

6 Work and Energy Principle

Work of a Force



Work of the force of gravity

$$dU = F_x dx + F_y dy + F_z dz$$

$$= -W dy$$

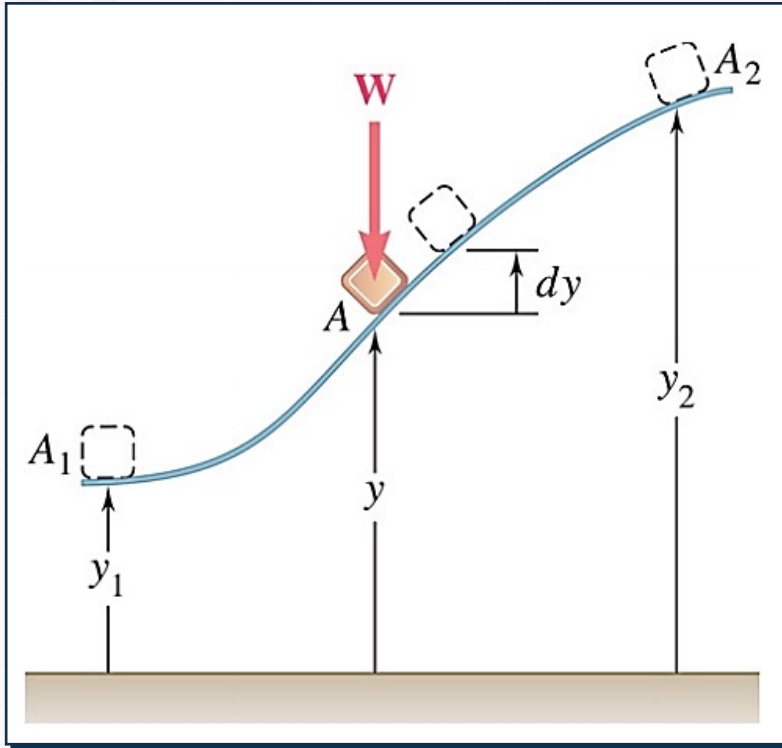
$$U_{1 \rightarrow 2} = - \int_{y_1}^{y_2} W dy$$

$$= -W(y_2 - y_1) = -W \Delta y$$

- Work of the weight is equal to product of weight W and vertical displacement Δy .

6 Work and Energy Principle

Work of a Force

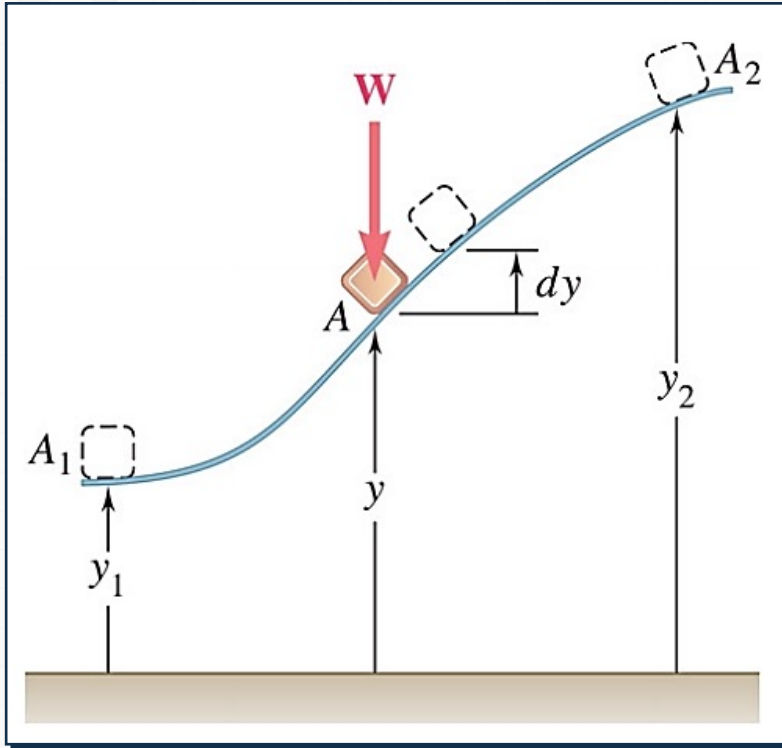


⚙ When is the work done by the weight positive?

- a) Moving from y_1 to y_2
- b) Moving from y_2 to y_1
- c) Never

6 Work and Energy Principle

Work of a Force



⚙ When is the work done by the weight positive?

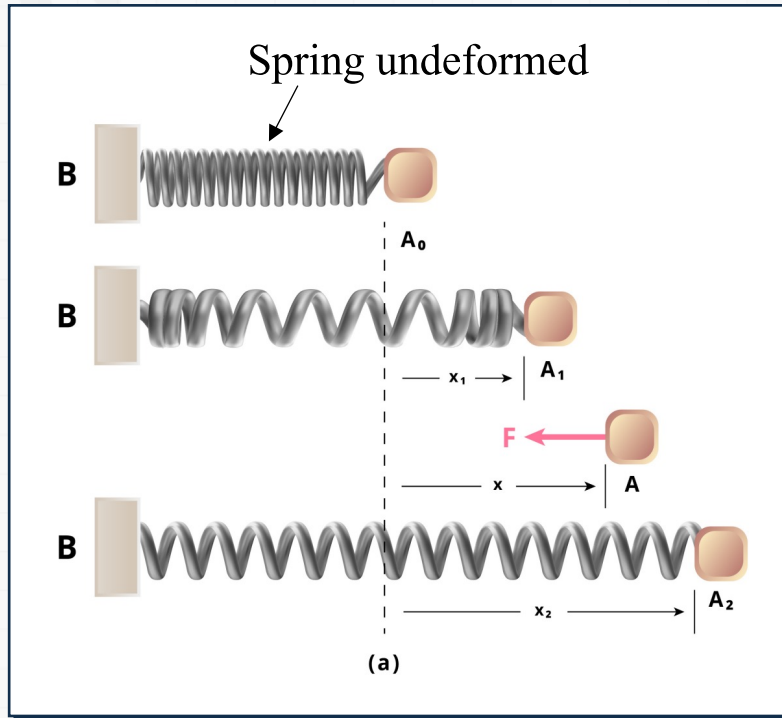
a) Moving from y_1 to y_2

b) Moving from y_2 to y_1

c) Never

6 Work and Energy Principle

Work of a Force



- Magnitude of the force exerted by a spring is proportional to deflection,

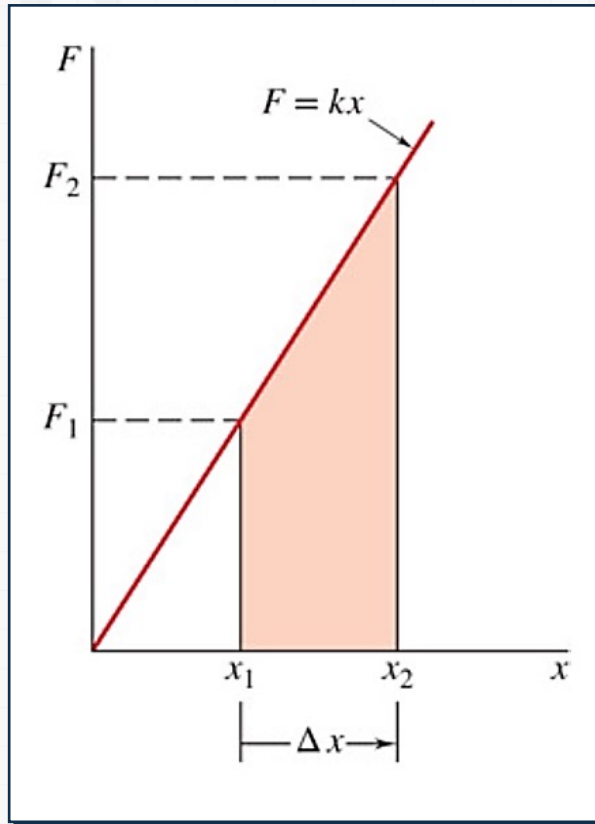
Work of the force exerted by spring,

$$dU = -F dx = -kx dx$$

$$U_{1 \rightarrow 2} = - \int_{x_1}^{x_2} kx dx = \frac{1}{2} kx_1^2 - \frac{1}{2} kx_2^2$$

6 Work and Energy Principle

Work of a Force



- Magnitude of the force exerted by a spring is proportional to deflection,

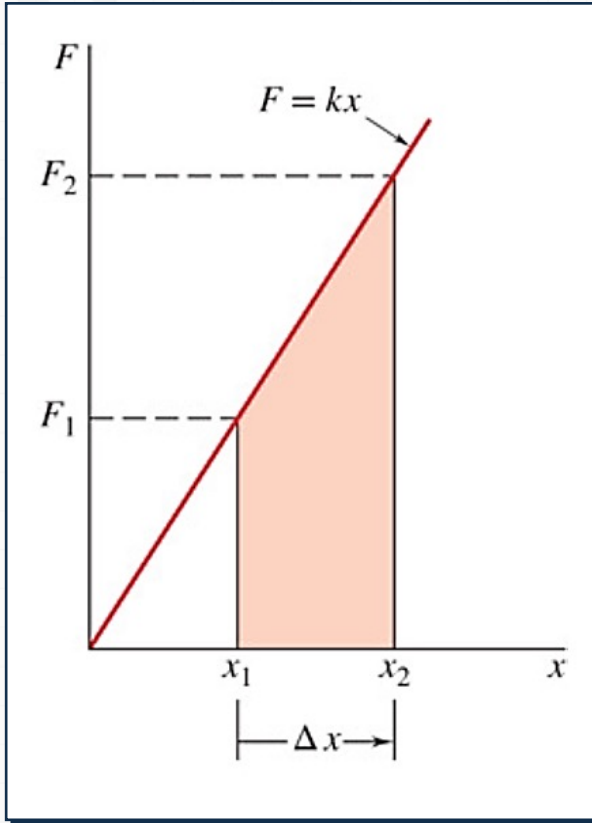
Work of the force exerted by spring,

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6 Work and Energy Principle

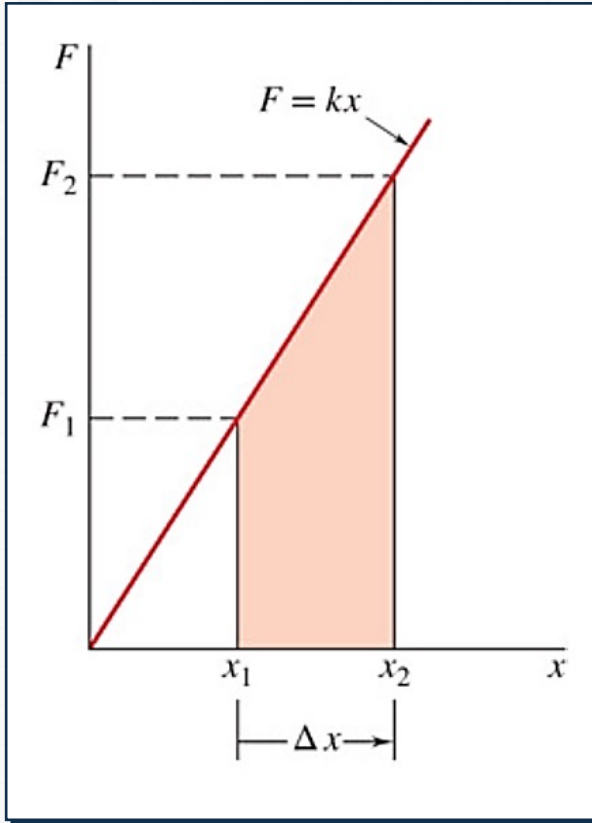
Work of a Force



- Work of the force exerted by spring is positive when $x_2 < x_1$
 - when the spring is returning to its undeformed position.

6 Work and Energy Principle

Work of a Force

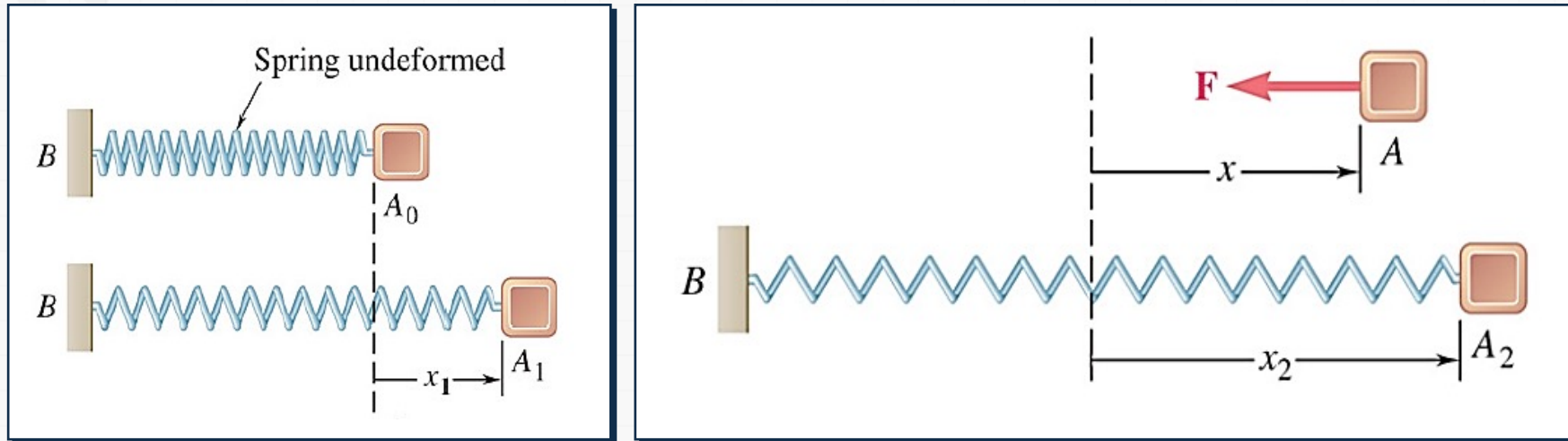


- Work of the force exerted by the spring is equal to negative of area under curve of F plotted against x ,

$$U_{1 \rightarrow 2} = -\frac{1}{2} (F_1 + F_2) \Delta x$$

6 Work and Energy Principle

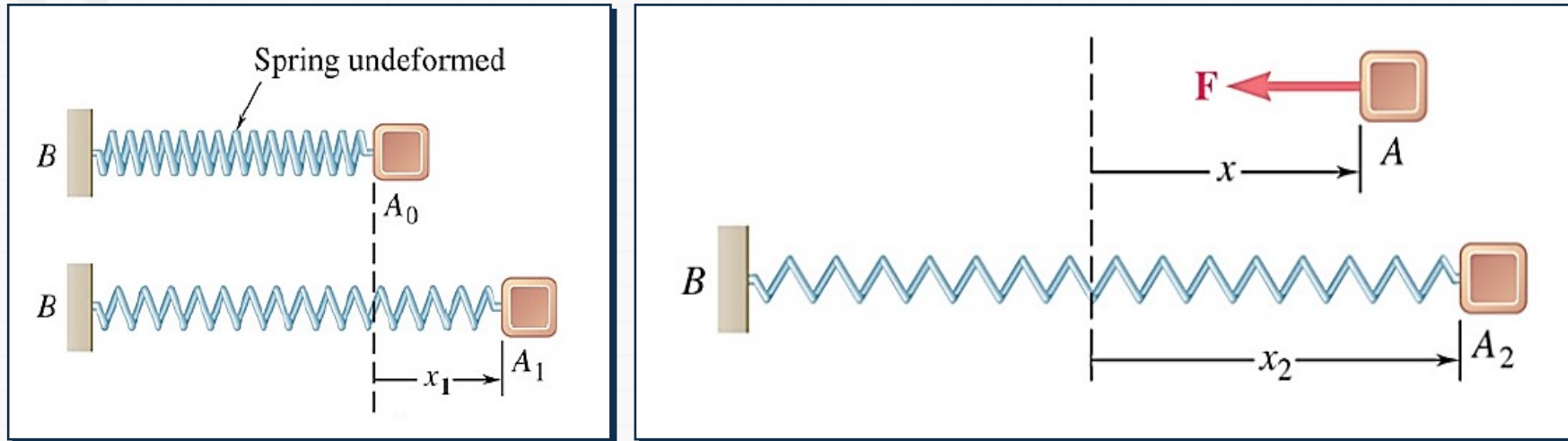
Work of a Force



➡ Displacement is in the opposite direction of the force.

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Work of a Force



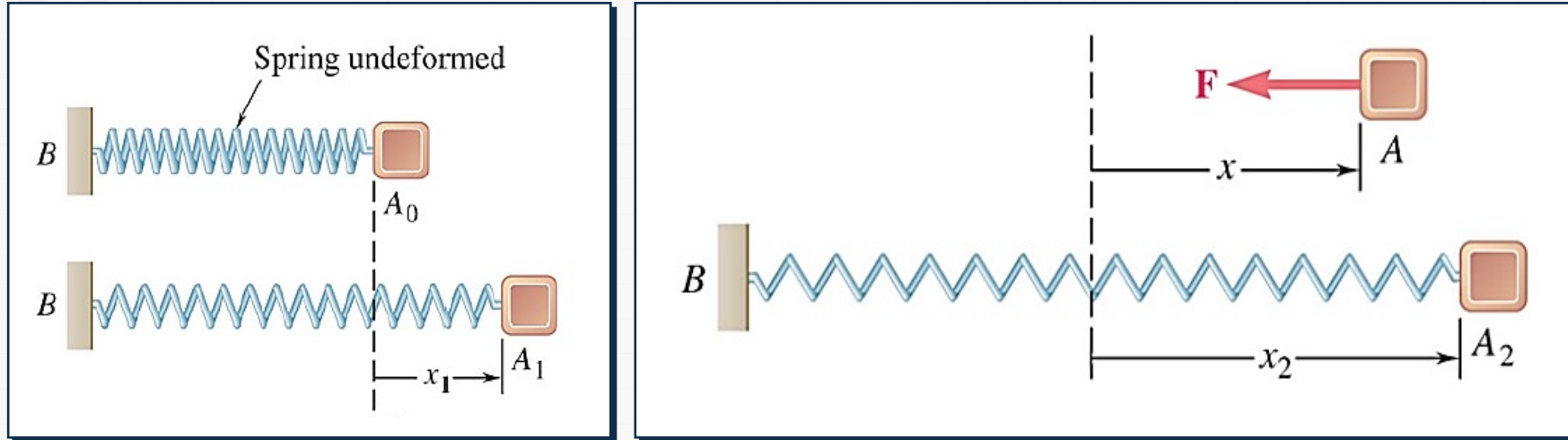
⚙️ As the block moves from A_0 to A_1 , is the work positive or negative?

a) Positive

b) Negative

6 Work and Energy Principle

Work of a Force



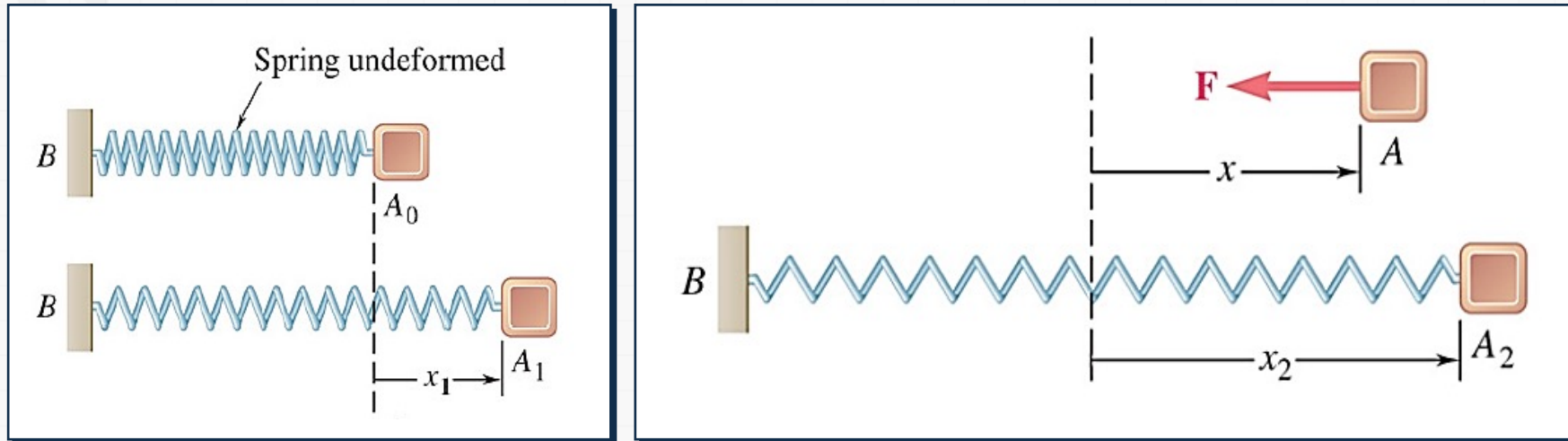
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6 Work and Energy Principle

Work of a Force



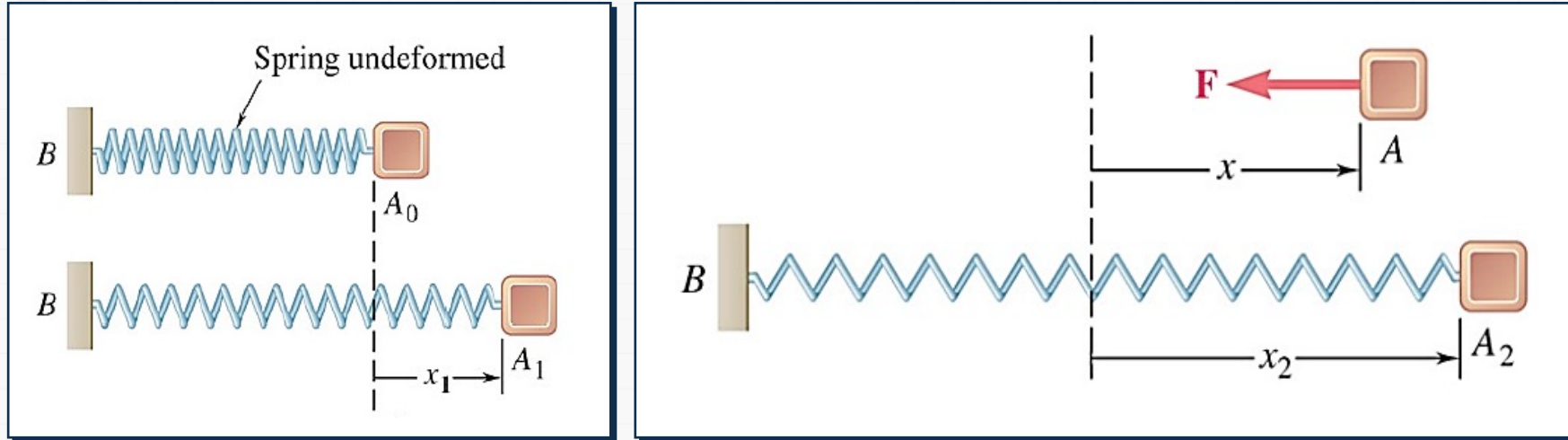
⚙️ As the block moves from A_2 to A_0 , is the work positive or negative?

a) Positive

b) Negative

6 Work and Energy Principle

Work of a Force



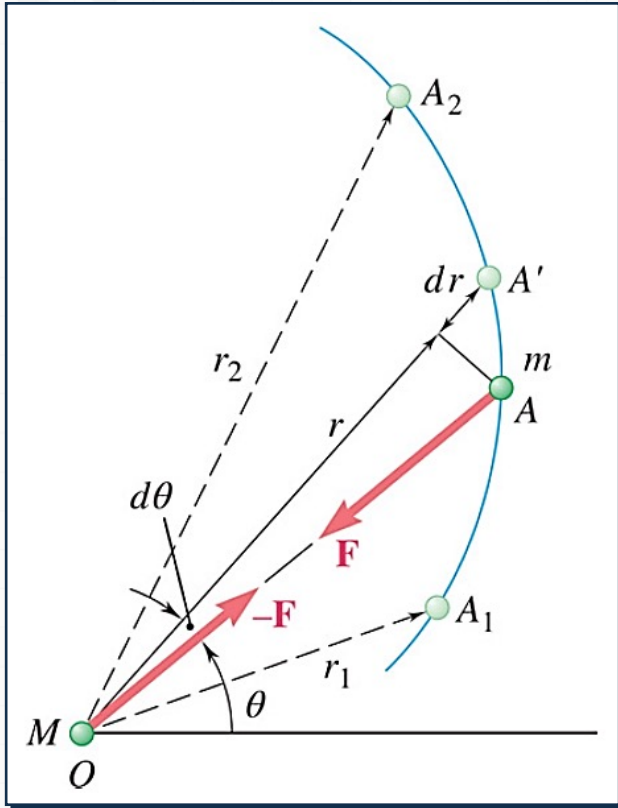
⚙️ As the block moves from A_2 to A_0 , is the work positive or negative?

a) Positive

b) Negative

6 Work and Energy Principle

Work of a Force



- Work of a gravitational force

(assume particle M occupies fixed position O while particle m follows path shown)

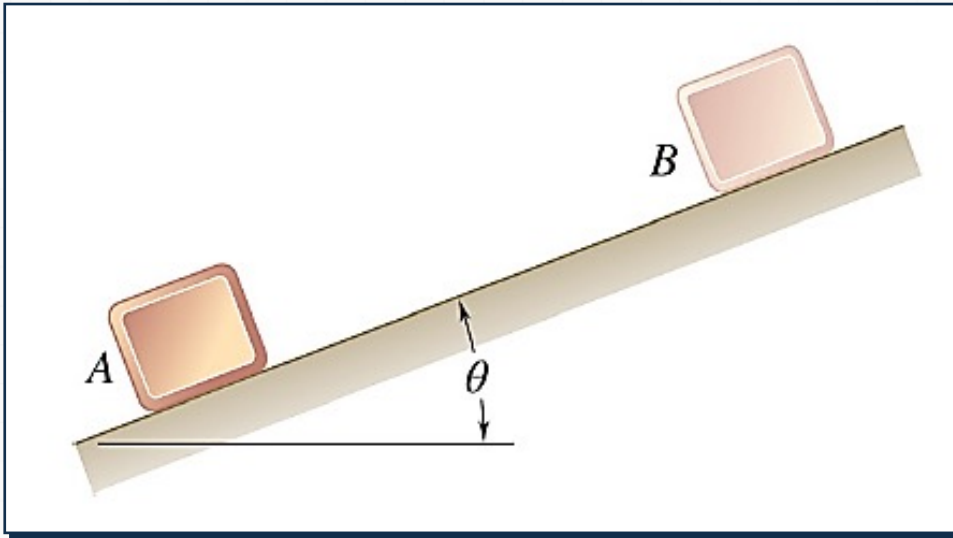
$$dU = -Fdr = -G \frac{Mm}{r^2} dr$$

$$U_{1 \rightarrow 2} = - \int_{r_1}^{r_2} G \frac{Mm}{r^2} dr = G \frac{Mm}{r_2} - G \frac{Mm}{r_1}$$

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Work of a Force

⚙ Does the normal force do work as the block slides from B to A?



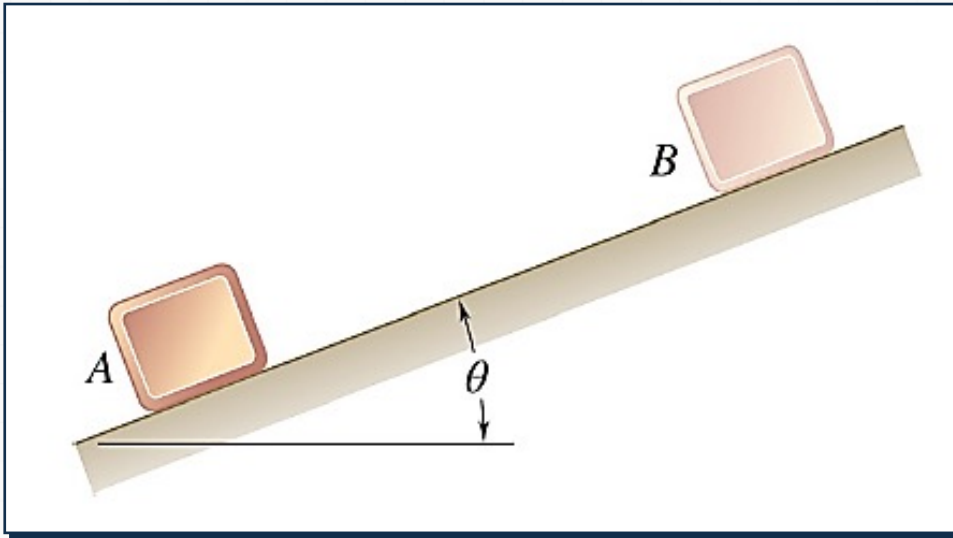
a) YES

b) NO

6 Work and Energy Principle

Work of a Force

⚙ Does the normal force do work as the block slides from B to A?



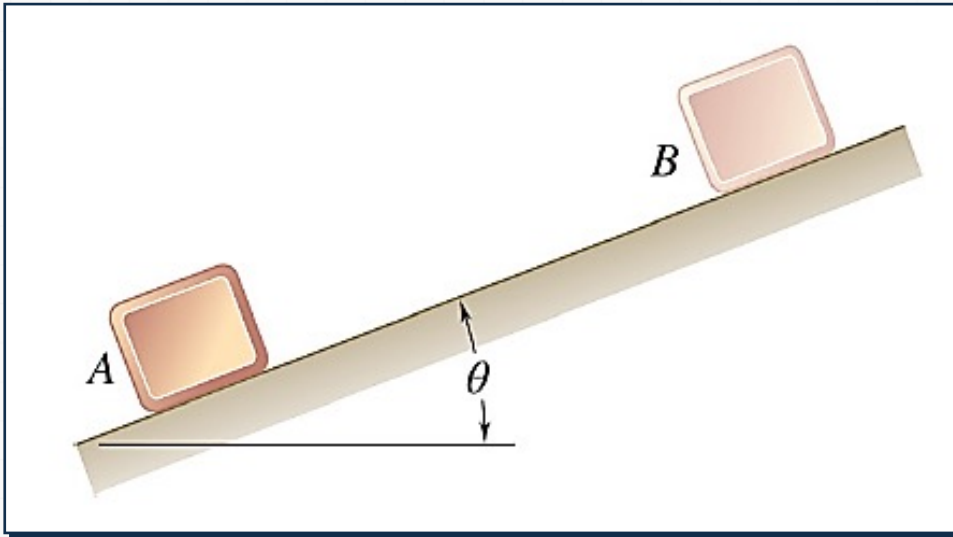
a) YES

b) NO

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Work of a Force

⚙ Does the weight do work as the block slides from B to A?



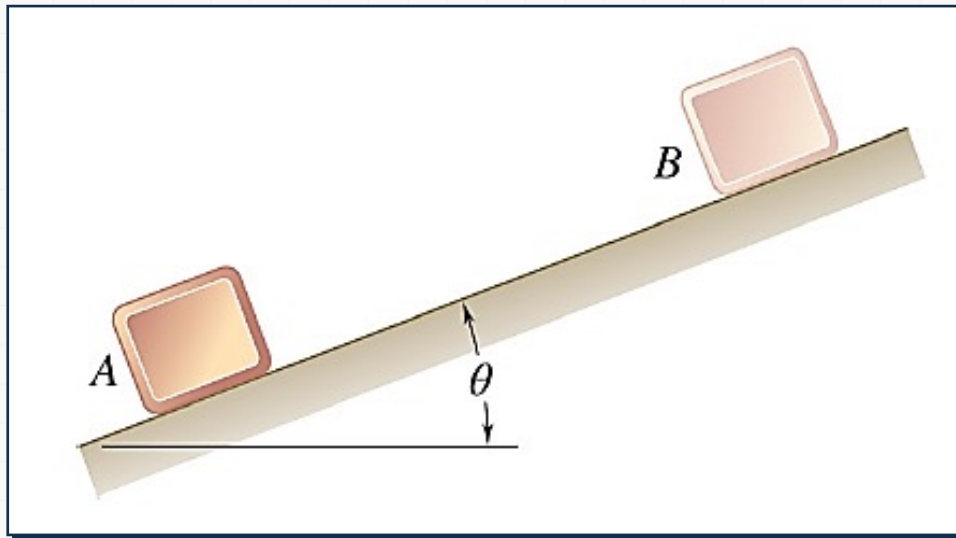
a) YES

b) NO

6 Work and Energy Principle

Work of a Force

⚙ Does the weight do work as the block slides from B to A?



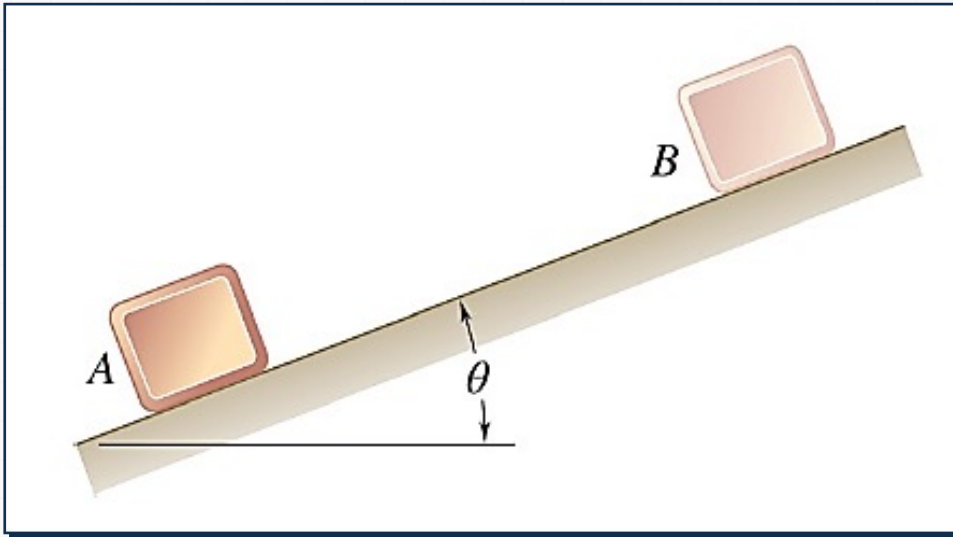
a) YES

b) NO

6 Work and Energy Principle

Work of a Force

⚙️ Positive or Negative work?



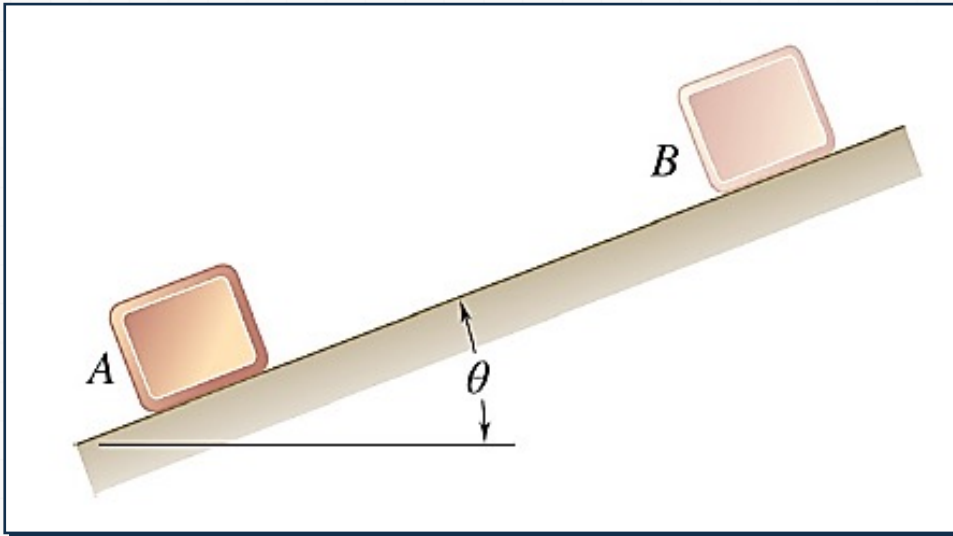
a) Positive

b) Negative

6 Work and Energy Principle

Work of a Force

⚙ Positive or Negative work?



a) Positive

b) Negative

6 Work and Energy Principle

Work of a Force

⚙️ **Forces which do not do work** ($ds = 0$ or $\cos a = 0$)

Reaction at frictionless pin supporting rotating body

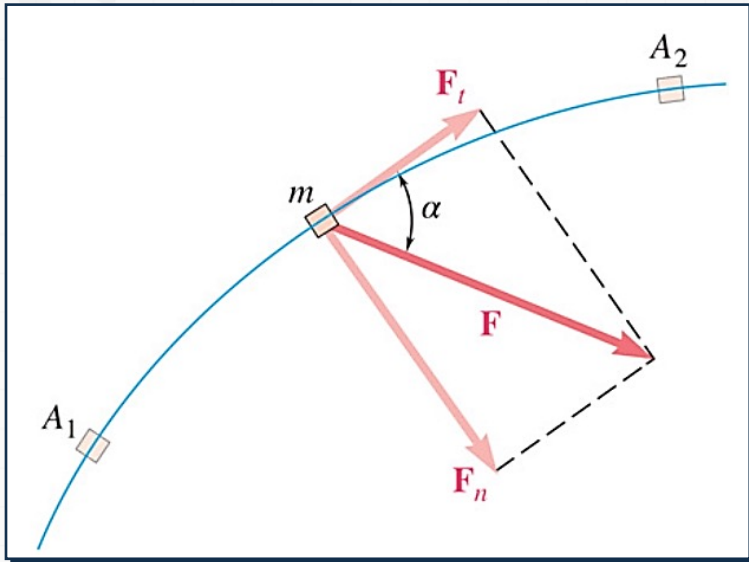
Reaction at frictionless surface when body in contact moves along surface

Reaction at a roller moving along its track

Weight of a body when its center of gravity moves horizontally

6 Work and Energy Principle

Principle of Work & Energy



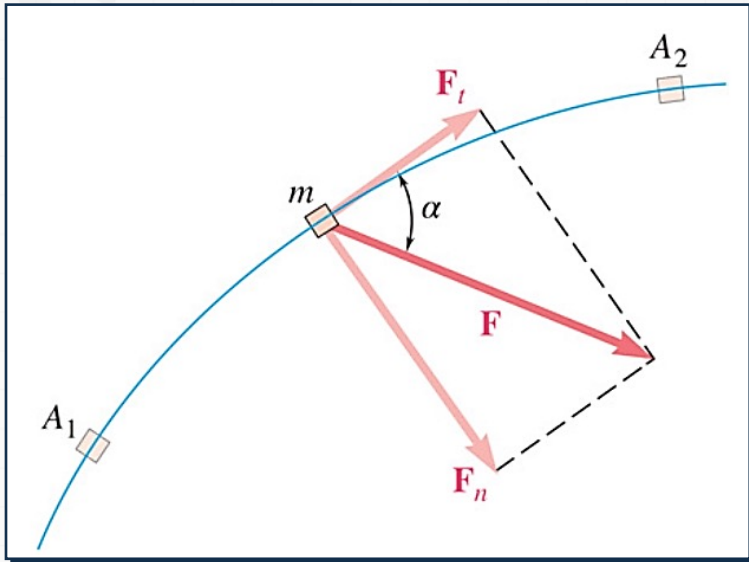
- Consider a particle of mass m acted upon by force $\vec{F}$,

$$\begin{aligned} F_t &= ma_t = m \frac{dv}{dt} \\ &= m \frac{dv}{ds} \frac{ds}{dt} = mv \frac{dv}{ds} \end{aligned}$$

$$F_t ds = mv dv$$

6 Work and Energy Principle

Principle of Work & Energy



• Integrating from A_1 to A_2 ,

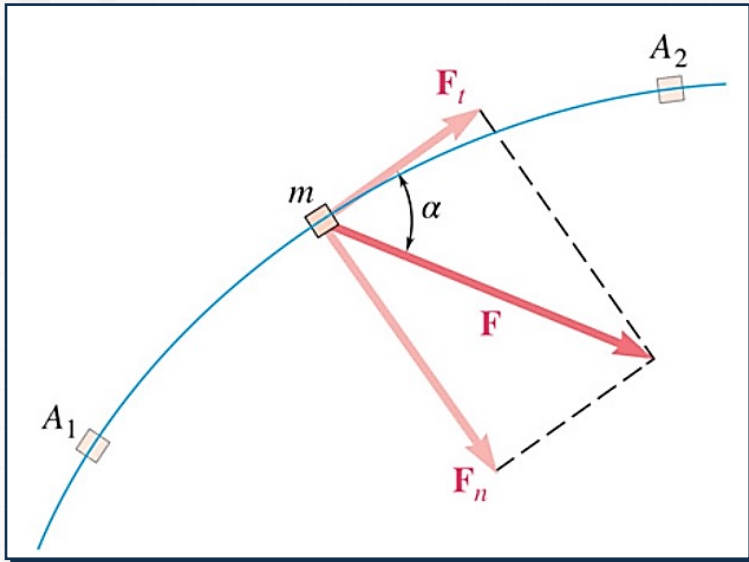
$$\int_{s_1}^{s_2} F_t ds = m \int_{v_1}^{v_2} v dv = \frac{1}{2} m v_1^2$$

$$U_{1 \rightarrow 2} = T_2 - T_1 \quad T = \frac{1}{2} m v^2 = \text{kinetic energy}$$

➡ The work of the force $\vec{F}$ is equal to the change in kinetic energy of the particle.

6 Work and Energy Principle

Principle of Work & Energy

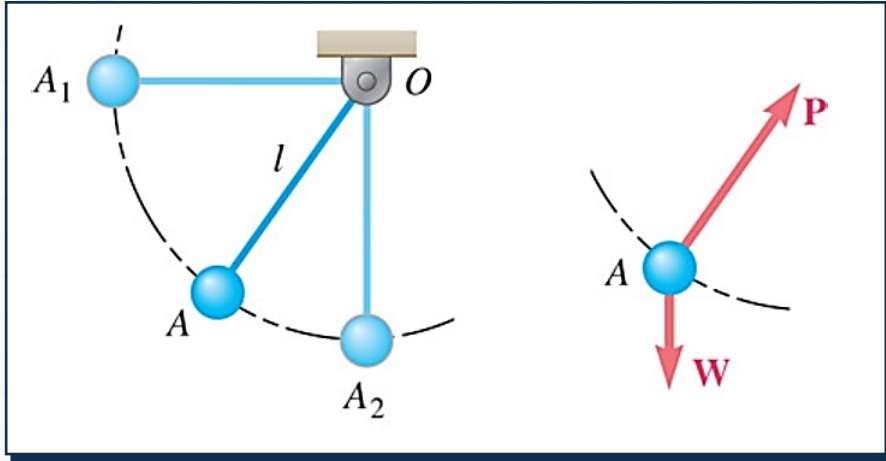


- Units of work and kinetic energy are the same.

$$T = \frac{1}{2}mv^2 = kg\left(\frac{m}{s}\right)^2 = \left(kg \frac{m}{s^2}\right)m = N \cdot m = J$$

6 Work and Energy Principle

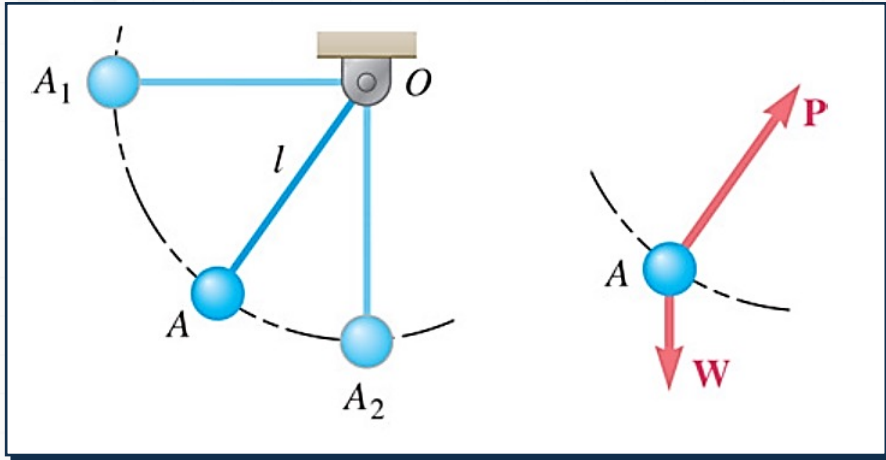
Applications of the Principle of Work and Energy



- The bob is released from rest at position A_1 .
 - It has magnitude and sign but not direction.

6 Work and Energy Principle

Applications of the Principle of Work and Energy



- Force $\vec{P}$ acts normal to path and does no work.

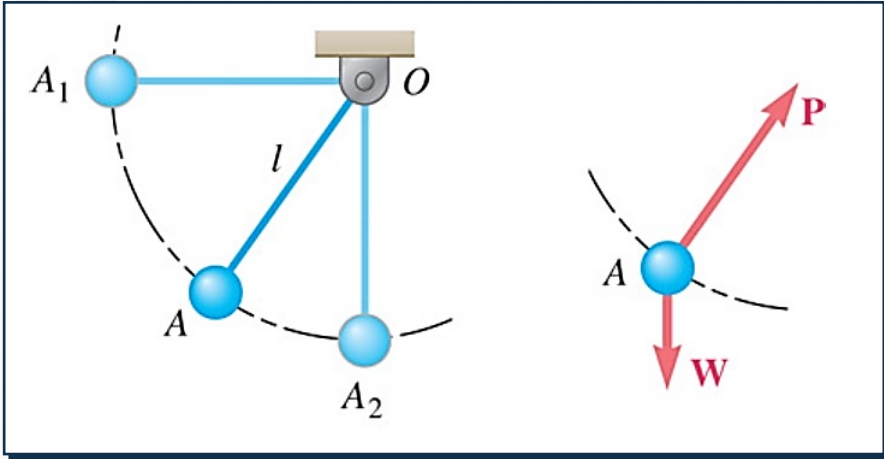
$$T_1 + U_{1 \rightarrow 2} = T_2$$

$$0 + Wl = \frac{1}{2} \frac{W}{g} v_2^2$$

$$v_2 = \sqrt{2gl}$$

6 Work and Energy Principle

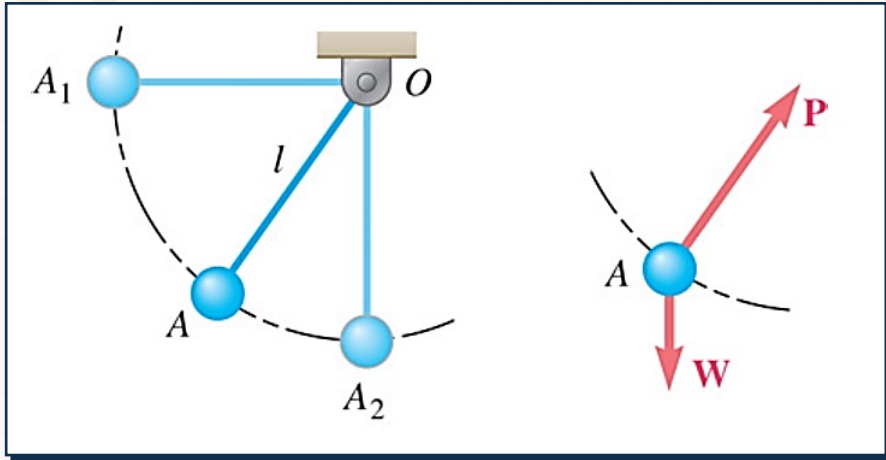
Applications of the Principle of Work and Energy



- Velocity is found without determining expression for acceleration and integrating.
- All quantities are scalars and can be added directly.
- Forces which do no work are eliminated from the problem.

6 Work and Energy Principle

Applications of the Principle of Work and Energy

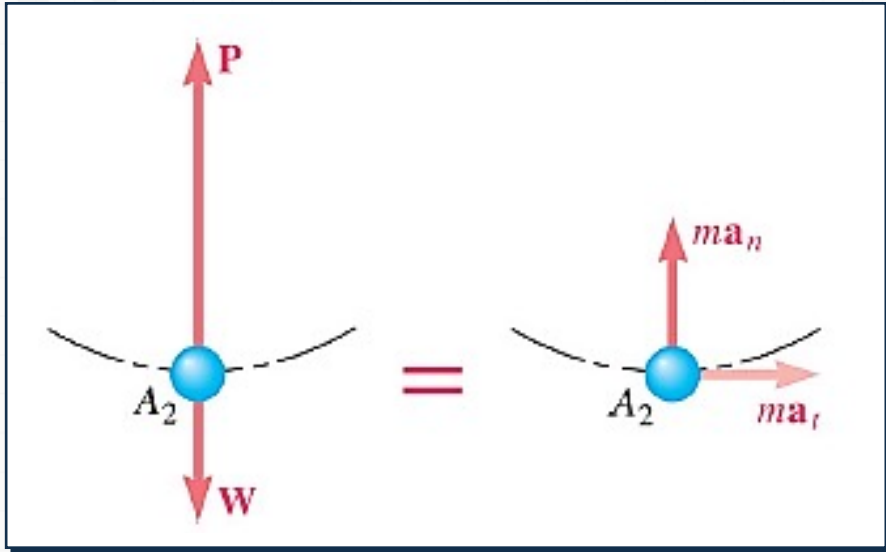


• Principle of Work and Energy

- Principle of work and energy cannot be applied to directly determine the acceleration of the pendulum bob.
- Calculating the tension in the cord requires supplementing the method of work and energy with an application of Newton's second law.

6 Work and Energy Principle

Applications of the Principle of Work and Energy



- As the bob passes through A_2 ,

$$\sum E_n = ma_n$$

$$P - W = \frac{W}{g} \frac{v_2^2}{l}$$

$$P = W + \frac{W}{g} \frac{2gl}{l} = 3W$$

$$v_2 = \sqrt{2gl}$$

➡ If you designed the rope to hold twice the weight of the bob, what would happen?

6 Work and Energy Principle

Power and Efficiency

⚙️ **Power = rate at which work is done.**

Power

$$= \frac{dU}{dt} = \frac{\vec{F} \cdot d\vec{r}}{dt} = \vec{F} \cdot \vec{v}$$

6 Work and Energy Principle

Power and Efficiency

⚙️ **Power = rate at which work is done.**

- Dimensions of power are work/time or force*velocity.
 - Units for power are:

$$1W(\text{watt}) = 1 \frac{J}{s} = 1N \cdot \frac{m}{s}$$

or

$$1hp = 550 \frac{ft \cdot lb}{s} = 746W$$

$$\eta = \text{efficiency}$$

$$= \frac{\text{output work}}{\text{input work}}$$

$$= \frac{\text{power output}}{\text{power input}}$$

6 Work and Energy Principle

Potential Energy

The potential energy stored at the top of the ball's path is transferred to kinetic energy as the ball meets the ground. Why is the ball's height reducing?



6 Work and Energy Principle

Potential Energy

If the work of a force only depends on differences in position, we can express this work as potential energy.

• Can the work done by the following forces be expressed as potential energy?

Weight	Yes	No
Friction	Yes	No
Normal force	Yes	No
Spring force	Yes	No

6 Work and Energy Principle

Potential Energy

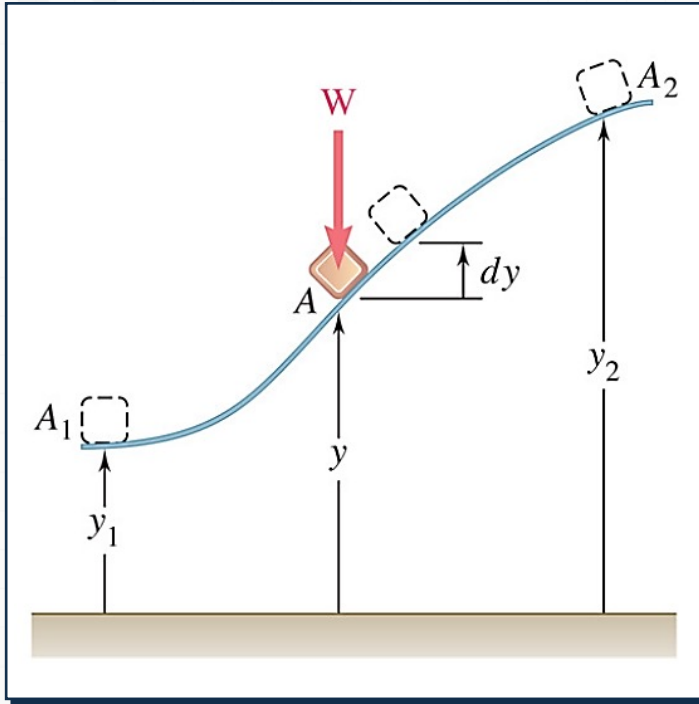
If the work of a force only depends on differences in position, we can express this work as potential energy.

• Can the work done by the following forces be expressed as potential energy?

Weight	Yes	No
Friction	Yes	No
Normal force	Yes	No
Spring force	Yes	No

6 Work and Energy Principle

Potential Energy



- Work of the force of gravity $\vec{W}$
 - Work is independent of path followed; depends only on the initial and final values of W_y .

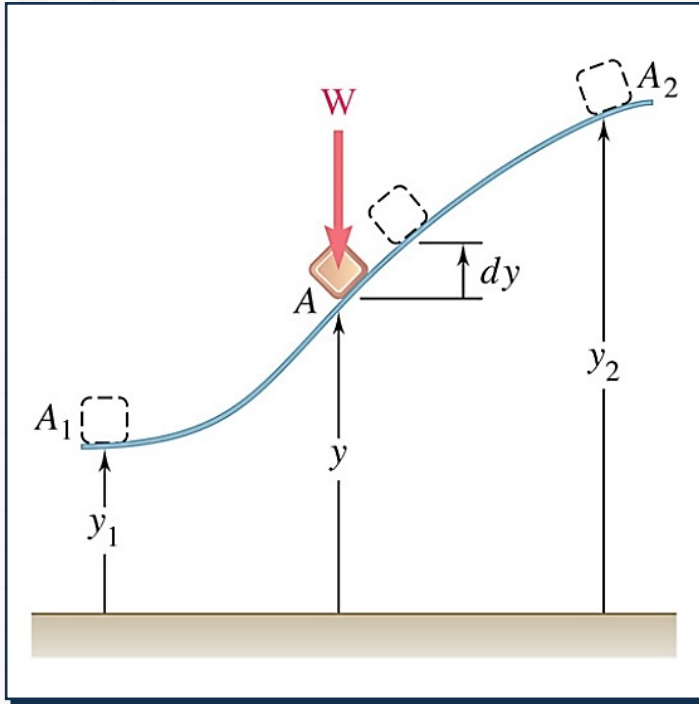
$$V_g = W_y$$

= *potential energy* of the body
with respect to *force of gravity*.

$$U_{1 \rightarrow 2} = (V_g)_1 - (V_g)_2$$

6 Work and Energy Principle

Potential Energy

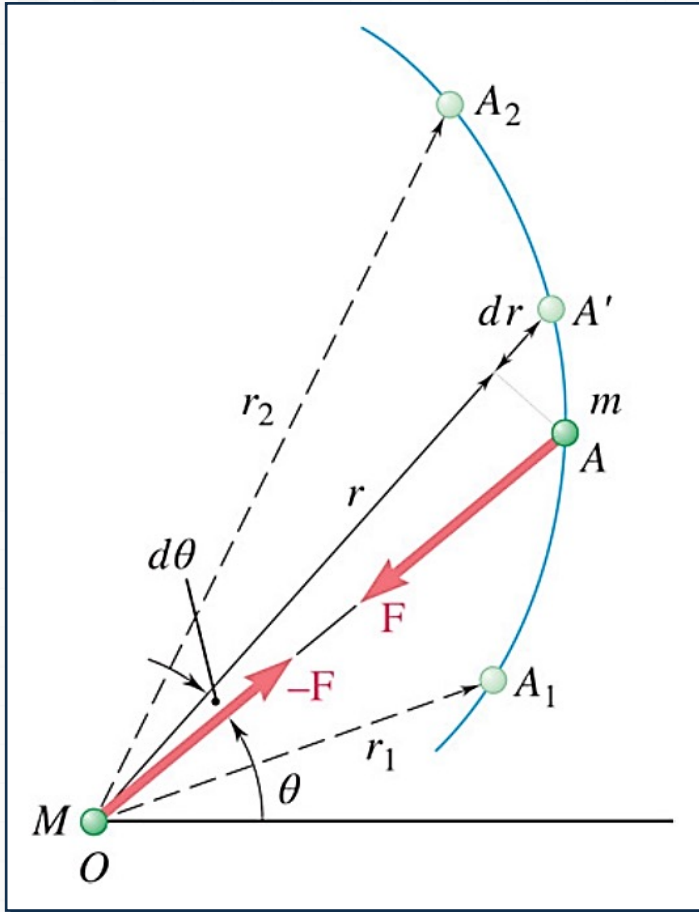


- Choice of datum from which the elevation y is measured is arbitrary.
- Units of work and potential energy are the same:

$$V_g = W_y = N \cdot m = J$$

6 Work and Energy Principle

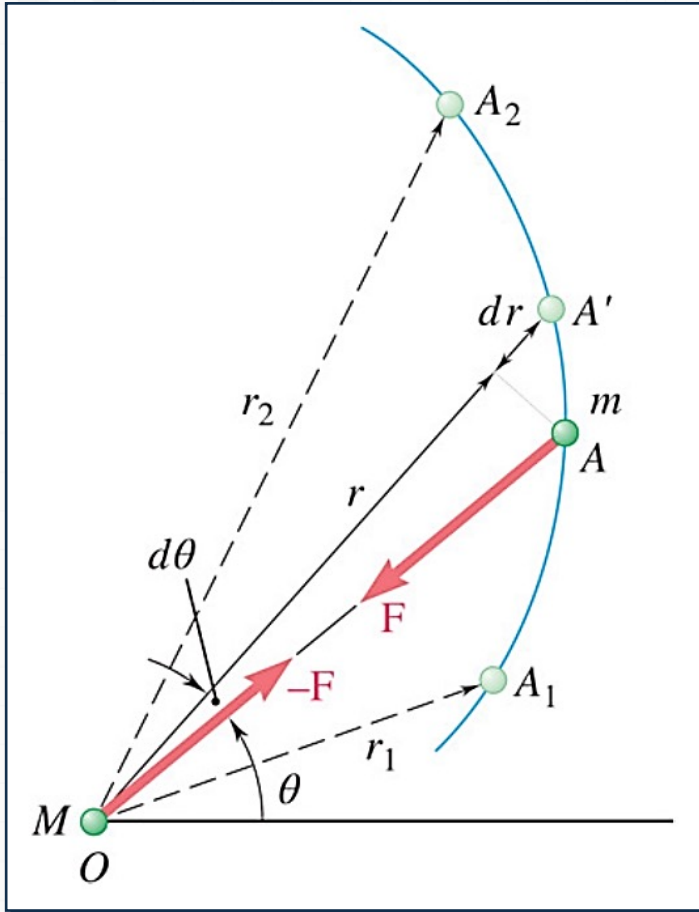
Potential Energy



- ➡ Previous expression for potential energy of a body with respect to gravity is only valid when the weight of the body can be assumed constant.
- ➡ For a space vehicle, the variation of the force of gravity with distance from the center of the earth should be considered.

6 Work and Energy Principle

Potential Energy



- Work of a gravitational force,

$$U_{1 \rightarrow 2} = \frac{GMm}{r_2} - \frac{GMm}{r_1}$$

- Potential energy V_g when the variation in the force of gravity can not be neglected,

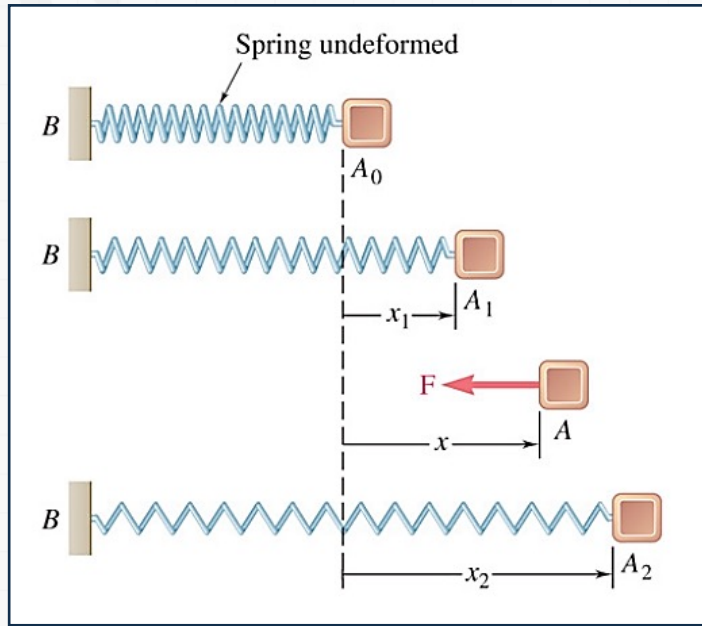
$$V_g = -\frac{GMm}{r} = -\frac{WR^2}{r}$$

W = Weight on surface of earth

R = radius of earth

6 Work and Energy Principle

Potential Energy

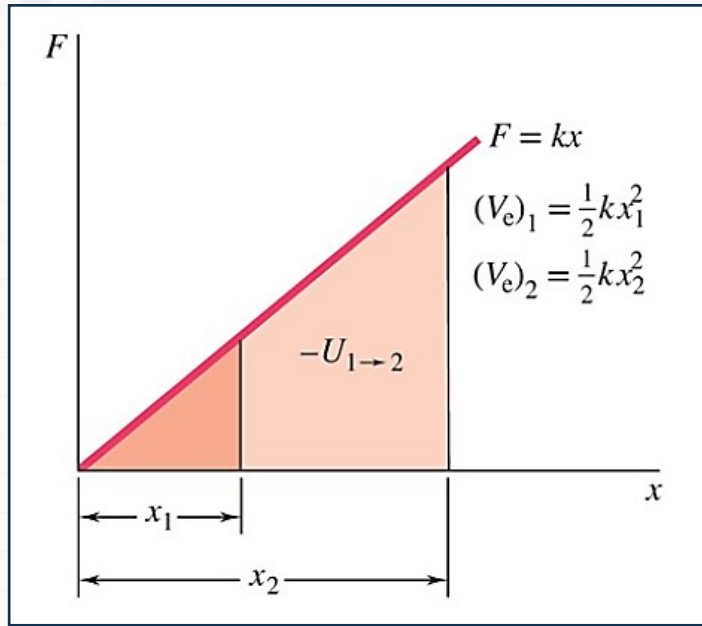


- Work of the force exerted by a spring depends only on the initial and final deflections of the spring,

$$U_{1 \rightarrow 2} = \frac{1}{2} k x_1^2 - \frac{1}{2} k x_2^2$$

6 Work and Energy Principle

Potential Energy



- The potential energy of the body with respect to the elastic force,

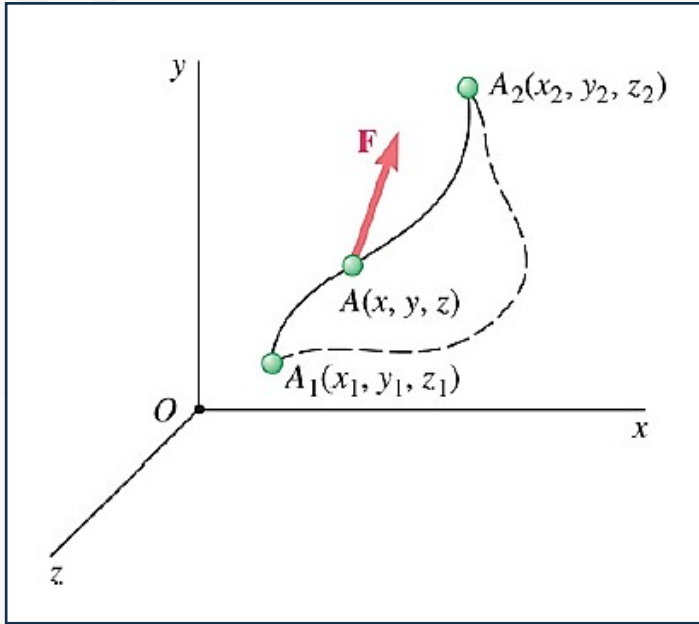
$$V_e = \frac{1}{2}kx^2$$

$$U_{1 \rightarrow 2} = (V_e)_1 - (V_e)_2$$

- Note that the preceding expression for V_e is valid only if the deflection of the spring is measured from its undeformed position.

6 Work and Energy Principle

Conservative Forces



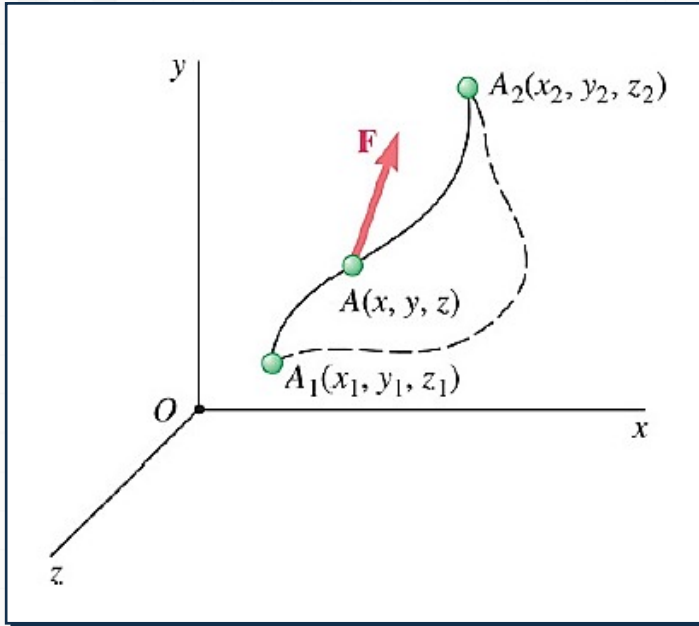
- Concept of potential energy can be applied
 - if the work of the force is independent of the path followed by its point of application.

$$U_{1 \rightarrow 2} = V(x_1, y_1, z_1) - V(x_2, y_2, z_2)$$

- Such forces are described as *conservative forces*.

6 Work and Energy Principle

Conservative Forces

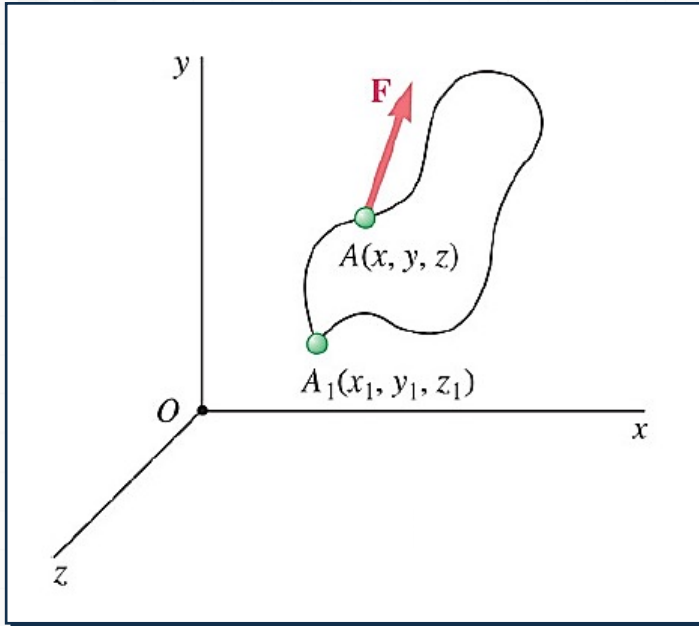


- For any conservative force applied on a closed path,

$$\oint \vec{F} \cdot d\vec{r} = 0$$

6 Work and Energy Principle

Conservative Forces



- Elementary work corresponding to displacement between two neighboring points,

$$dU = V(x, y, z) - V(x + dx, y + dy, z + dz)$$

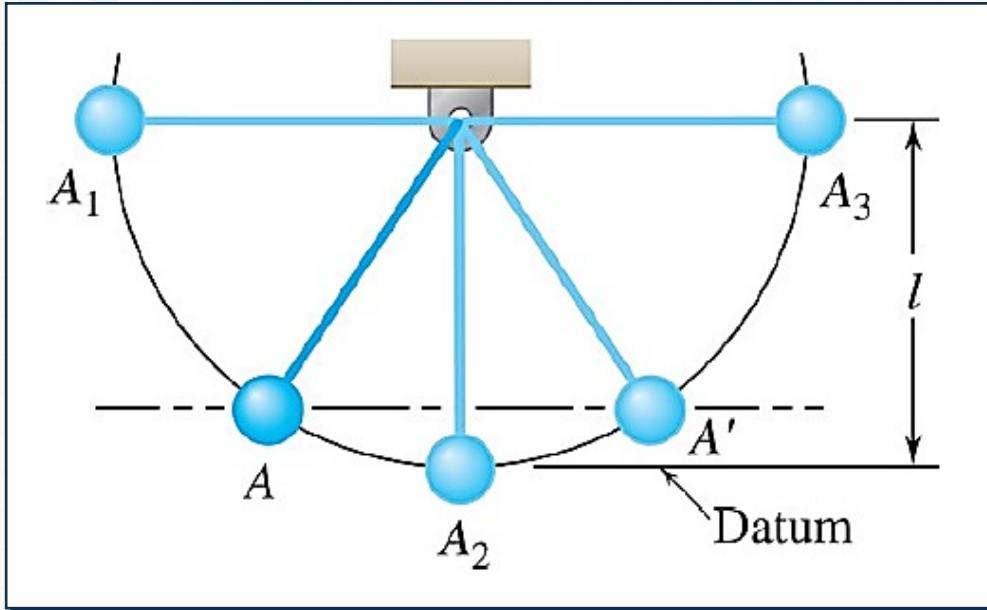
$$= -dV(x, y, z)$$

$$F_x dx + F_y dy + F_z dz = -\left(\frac{\partial V}{\partial x} dx + \frac{\partial V}{\partial y} dy + \frac{\partial V}{\partial z} dz\right)$$

$$\vec{F} = -\left(\frac{\partial V}{\partial x} + \frac{\partial V}{\partial y} + \frac{\partial V}{\partial z}\right) = -\nabla V$$

6 Work and Energy Principle

Conservation of Energy

**Work of a conservative force**

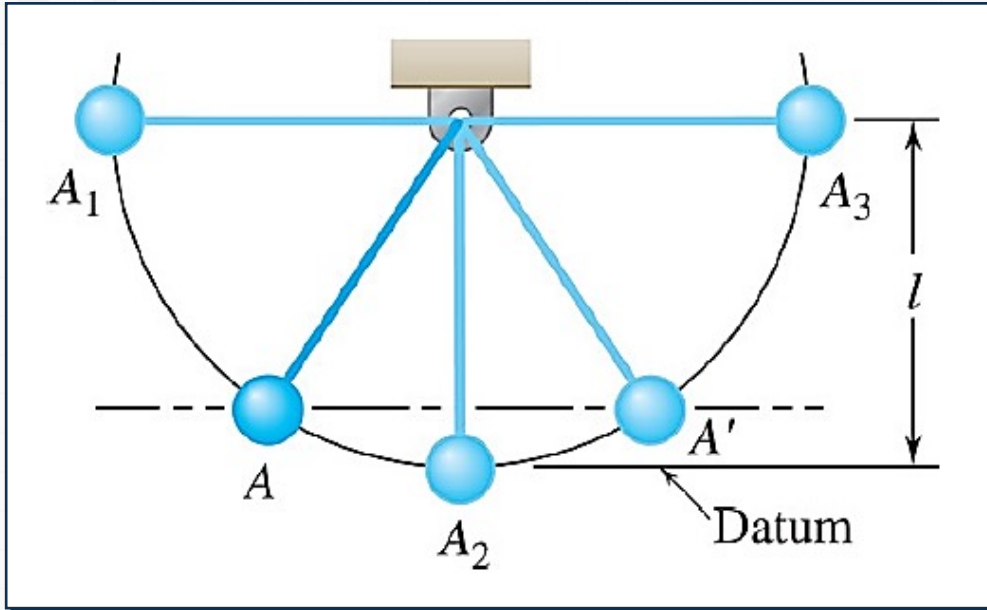
$$U_{1 \rightarrow 2} = V_1 - V_2$$

Concept of work and energy

$$U_{1 \rightarrow 2} = T_2 - T_1$$

6 Work and Energy Principle

Conservation of Energy



- Follows that

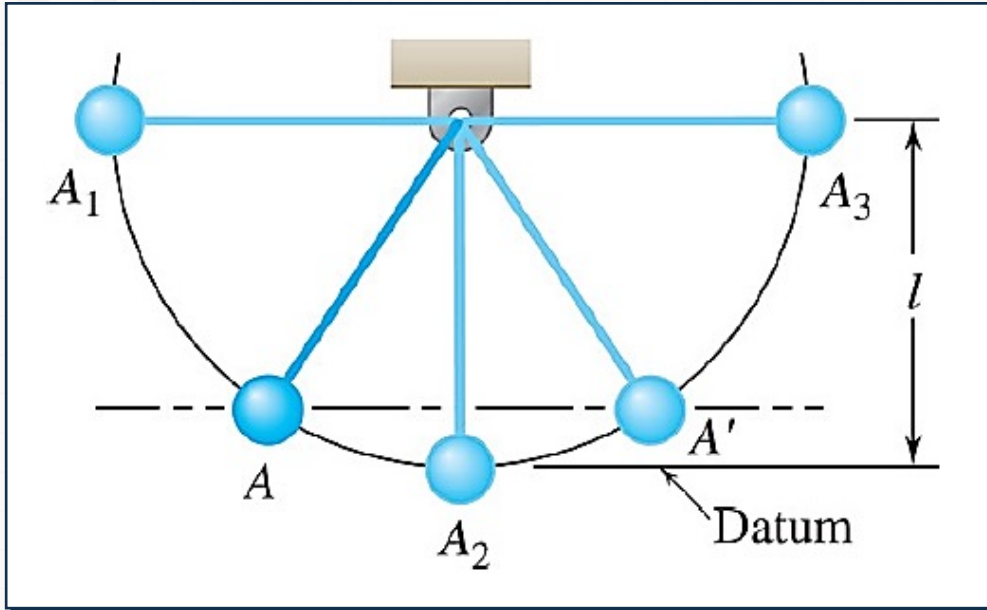
$$T_1 + V_1 = T_2 + V_2$$

$$E = T + V = \text{constant}$$

- When a particle moves under the action of conservative forces
 - The total mechanical energy is constant.

6 Work and Energy Principle

Conservation of Energy



- Friction forces are not conservative.
 - Total mechanical energy of a system involving friction decreases.
- Mechanical energy is dissipated by friction into thermal energy.
 - Total energy is constant

6 Work and Energy Principle

Conservation of Energy

$$T_1 = 0$$

$$T_1 = 0 \quad V_1 = W\ell$$

$$T_1 + V_1 = W\ell$$

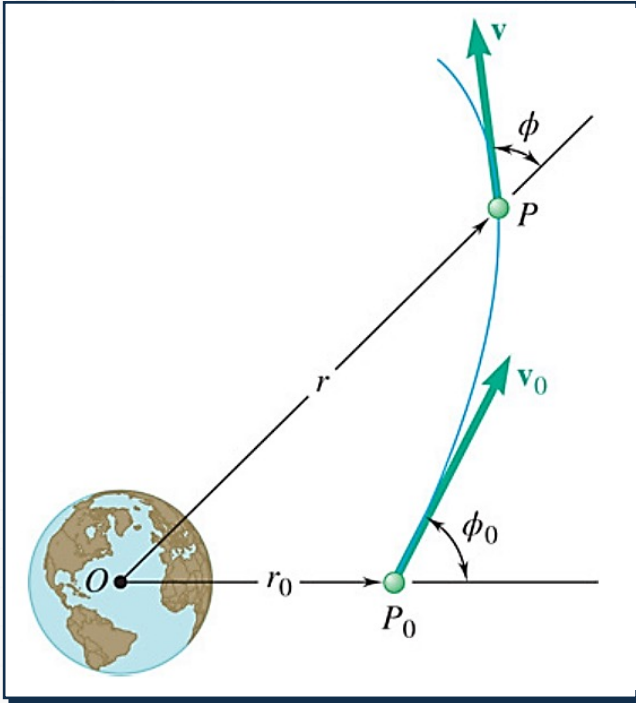
$$T_2 = \frac{1}{2}mv_2^2 = \frac{1}{2}\frac{W}{g}(2g\ell) = W\ell$$

$$V_0 = 0$$

$$T_2 + V_2 = W\ell$$

6 Work and Energy Principle

Motion Under a Conservative Central Force

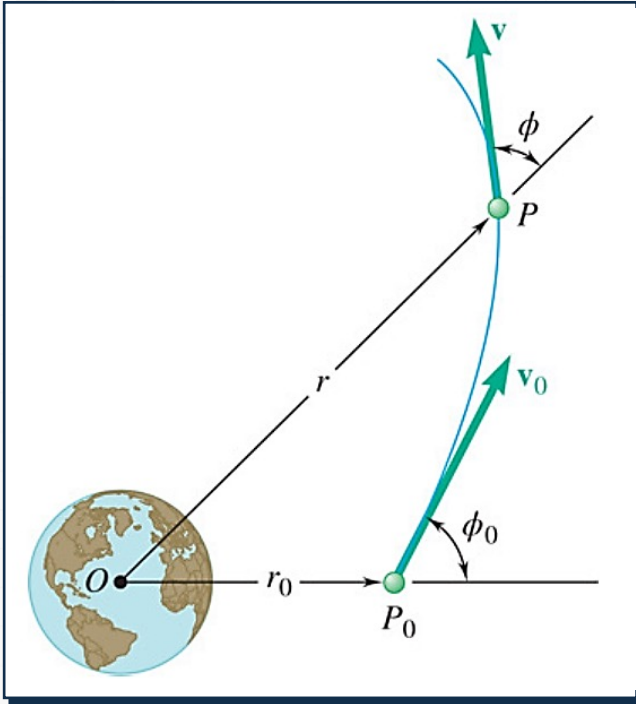


- When a particle moves under a conservative central force
 - Both the principle of conservation of angular momentum

$$r_0 m v_0 \sin \phi = r m v \sin \phi$$

6 Work and Energy Principle

Motion Under a Conservative Central Force



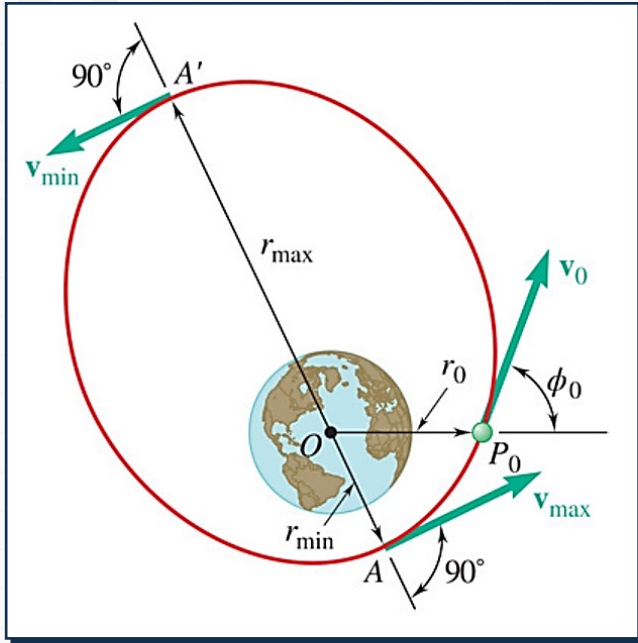
- When a particle moves under a conservative central force
 - The principle of conservation of energy may be applied.

$$T_0 + V_0 = T + V$$

$$\frac{1}{2} m v_0^2 - \frac{GMm}{r_0} = \frac{1}{2} m v^2 - \frac{GMm}{r}$$

6 Work and Energy Principle

Motion Under a Conservative Central Force

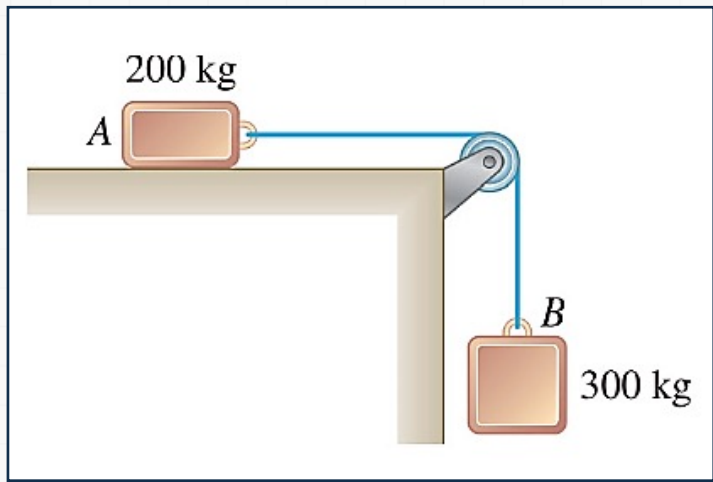


- Given r , the equations may be solved for v and φ .
- At minimum and maximum r , $\varphi = 90^\circ$
 - Given the launch conditions, the equations may be solved for $r_{\min}$, $r_{\max}$, $v_{\min}$ and $v_{\max}$.

6 Work and Energy Principle

Sample Problem 1

⚙️ Two blocks are joined by an inextensible cable as shown.



• Determine

- If the system is released from rest, determine the velocity of block A after it has moved $2m$.
- Assume that the coefficient of friction between block A and the plane is $\mu_k = 0.25$ and that the pulley is weightless and frictionless.

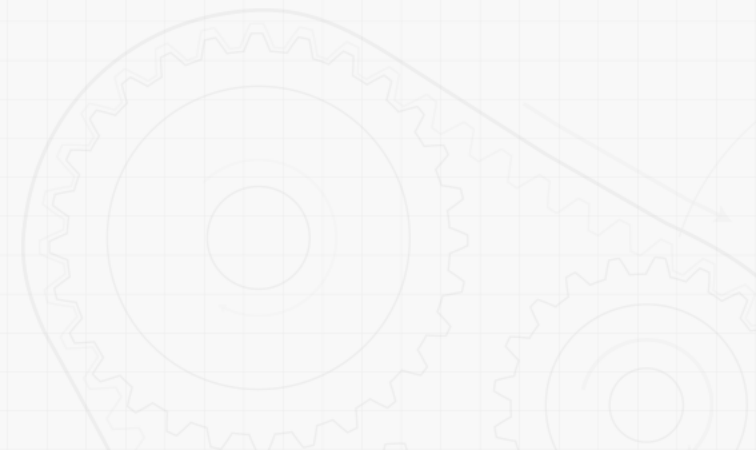
6 Work and Energy Principle

Sample Problem 1

⚙️ **Two blocks are joined by an inextensible cable as shown.**

• **Strategy**

- Apply the principle of work and energy separately to blocks A and B .
- When the two relations are combined, the work of the cable forces cancel. Solve for the velocity.

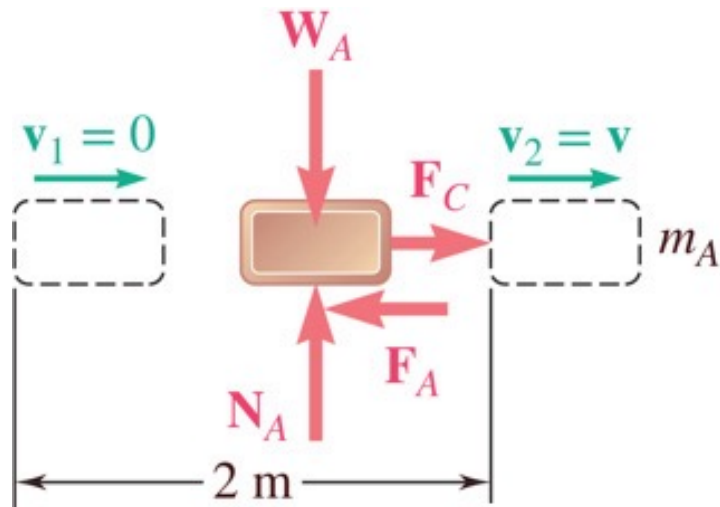


6 Work and Energy Principle

Sample Problem 1

⚙ Modeling and Analysis

- Apply the principle of work and energy separately to blocks A and B .



$$W_A = (200\text{ kg})(9.81\text{ m/s}^2) = 1962\text{ N}$$

$$F_A = \mu_k N_A = \mu_k W_A = 0.25(1962\text{ N}) = 490\text{ N}$$

$$T_1 + U_{1 \rightarrow 2} = T_2:$$

$$0 + F_C(2) - F_A(2) = \frac{1}{2} m_A v^2$$

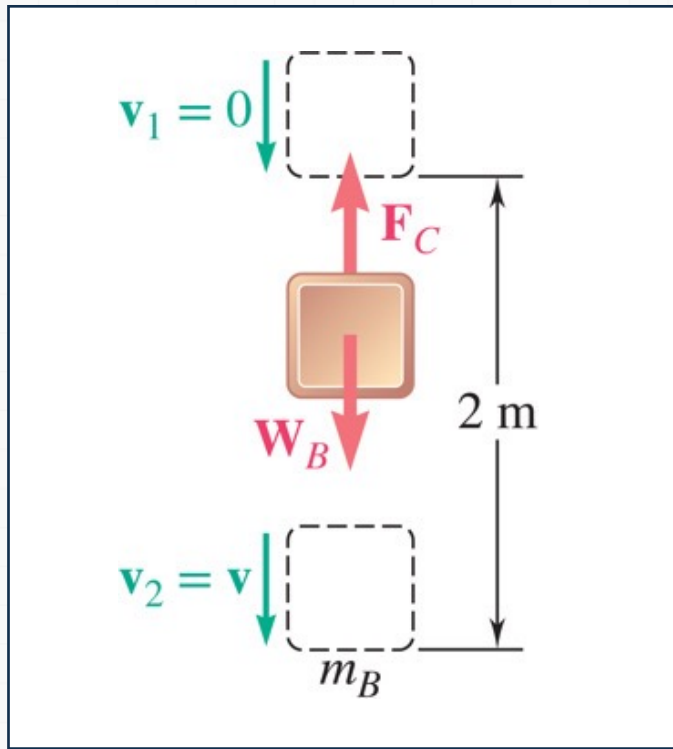
$$F_C(2) - (490)(2) = \frac{1}{2} (200) v^2$$

6 Work and Energy Principle

Sample Problem 1

⚙ Modeling and Analysis

- Apply the principle of work and energy separately to blocks *A* and *B*.



$$W_B = (300\text{kg})(9.81\text{m/s}^2) = 2960\text{N}$$

$$T_1 + U_{1 \rightarrow 2} = T_2:$$

$$0 - F_C(2) + W_B(2) = \frac{1}{2}m_B v^2$$

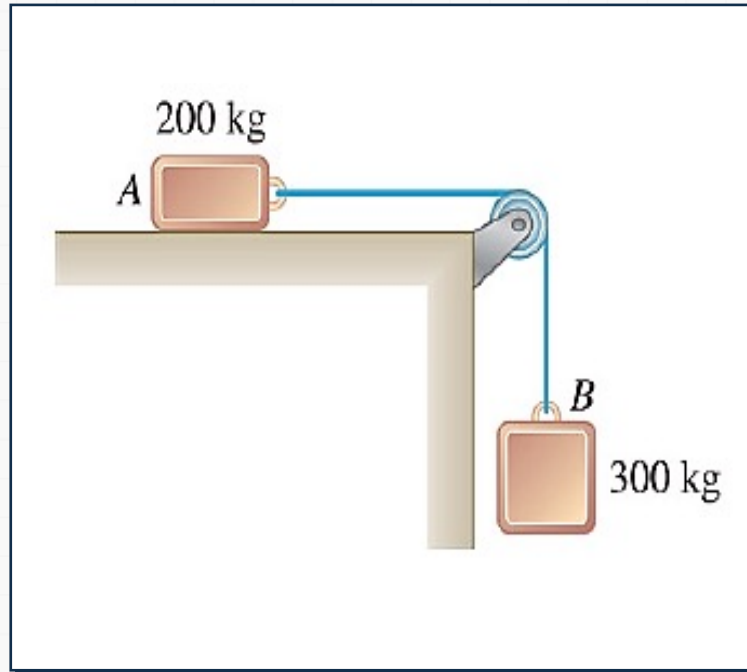
$$-F_C(2) - (2940)(2) = \frac{1}{2}(300)v^2$$

6 Work and Energy Principle

Sample Problem 1

⚙ Modeling and Analysis

- When the two relations are combined, the work of the cable forces cancel.
 - Solve for the velocity.



$$F_C(2) - (490)(2) = \frac{1}{2}(200)v^2$$

$$-F_C(2) + (2940)(2) = \frac{1}{2}(300)v^2$$

$$(2940)(2) - (490)(2) = \frac{1}{2}(200 + 300)v^2$$

$$4900 = \frac{1}{2}(500)v^2$$

$$v = 4.43 \text{ m/s}$$

6 Work and Energy Principle

Sample Problem 1

⚙️ Reflect and Think

This problem can also be solved by applying the principle of work and energy to the combined system of blocks.

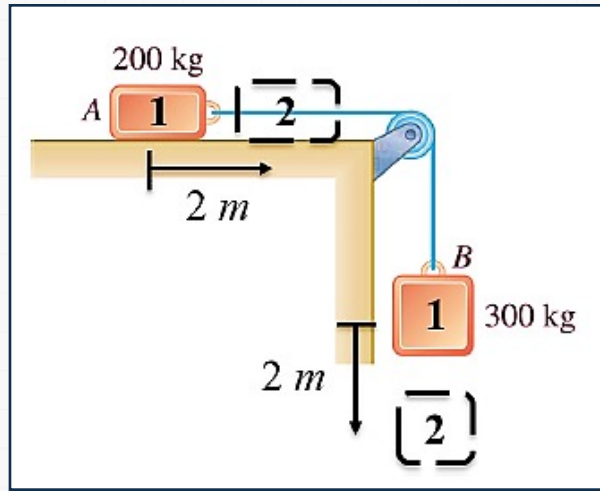
When using the principle of work and energy, it usually saves time to choose your system to be everything that moves.

6 Work and Energy Principle

Sample Problem 1

⚙️ Alternate Solution, Group Problem Solving

- Could you apply work-energy to the combined system of blocks?



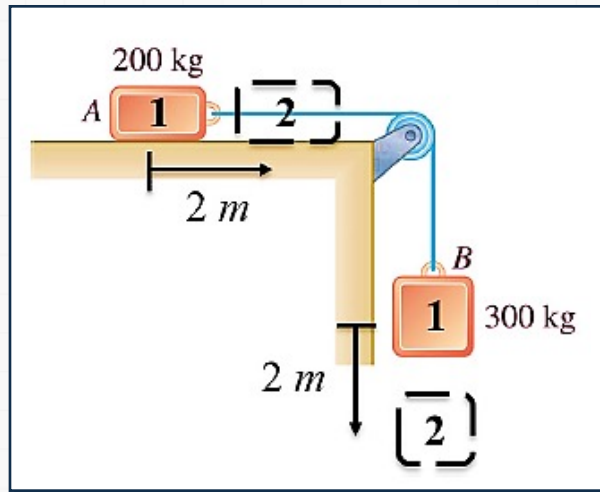
Given: $v_1 = 0$, distance = 2 m, $\mu_k = 0.25$

6 Work and Energy Principle

Sample Problem 1

⚙️ Alternate Solution, Group Problem Solving

- Could you apply work-energy to the combined system of blocks?



- What is T_1 of the system?

$$T_1 = 0$$

- What is the total work done between points 1 and 2?

$$\begin{aligned} U_{1 \rightarrow 2} &= -(0.25)(200)(9.81)(2) + (300)(9.81)(2) \\ &= 4900J \end{aligned}$$

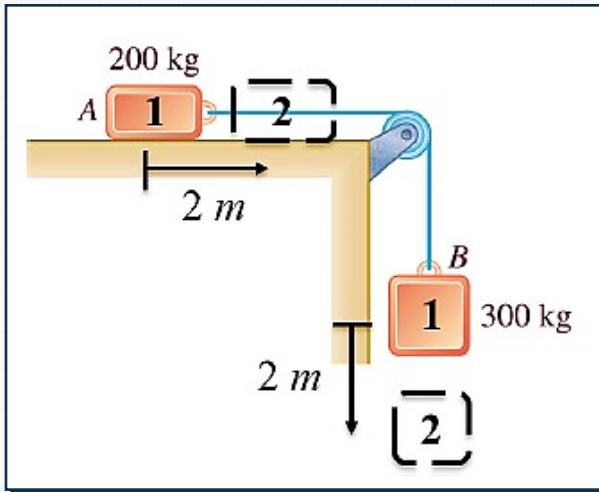
➡ Note that $v_A = v_B$

6 Work and Energy Principle

Sample Problem 1

⚙️ Alternate Solution, Group Problem Solving

- Could you apply work-energy to the combined system of blocks?



- What is T_2 of the system?

$$T_2 = \frac{1}{2}m_A v^2 + \frac{1}{2}m_B v^2 = \frac{1}{2}(200)v^2 + \frac{1}{2}(300)v^2$$

- What is the total work done between points 1 and 2?

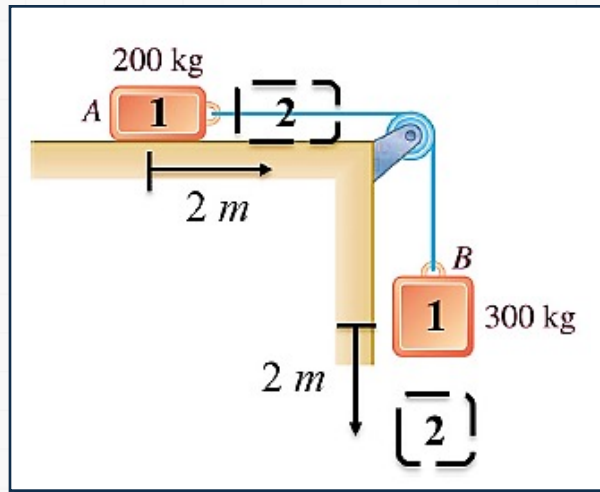
$$T_2 = \frac{1}{2}m_A v^2 + \frac{1}{2}m_B v^2 = \frac{1}{2}(200)v^2 + \frac{1}{2}(300)v^2$$

6 Work and Energy Principle

Sample Problem 1

⚙️ Alternate Solution, Group Problem Solving

- Could you apply work-energy to the combined system of blocks?



- Solve for v

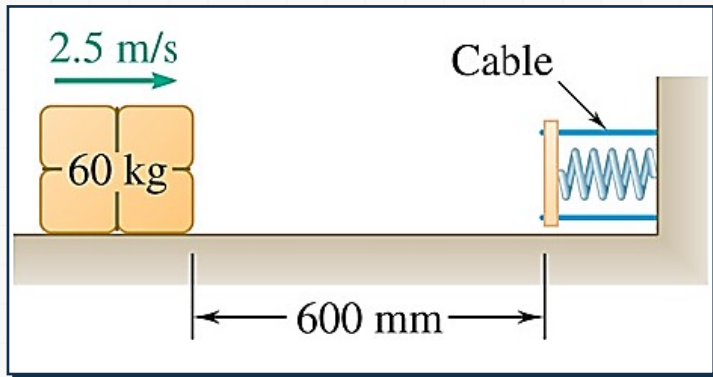
$$4900 = \frac{1}{2} (500) v^2$$

$$v = 4.43 \text{ m/s}$$

6 Work and Energy Principle

Sample Problem 2

⚙️ A spring is used to stop a 60 kg package which is sliding on a horizontal surface.



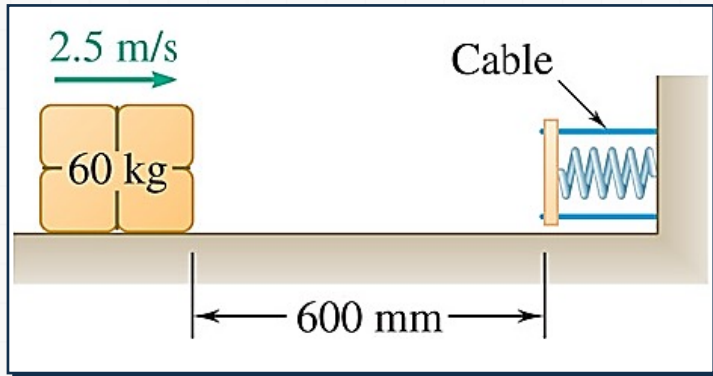
• Determine

- The spring has a constant $k = 20\text{ kN/m}$ and is held by cables so that it is initially compressed 120 mm .
- The package has a velocity of 2.5 m/s in the position shown and the maximum deflection of the spring is 40 mm .

6 Work and Energy Principle

Sample Problem 2

⚙️ A spring is used to stop a 60 kg package which is sliding on a horizontal surface.



• Determine

- Determine (a) the coefficient of kinetic friction between the package and surface and (b) the velocity of the package as it passes again through the position shown.

6 Work and Energy Principle

Sample Problem 2

⚙️ **A spring is used to stop a 60kg package which is sliding on a horizontal surface.**

• **Strategy**

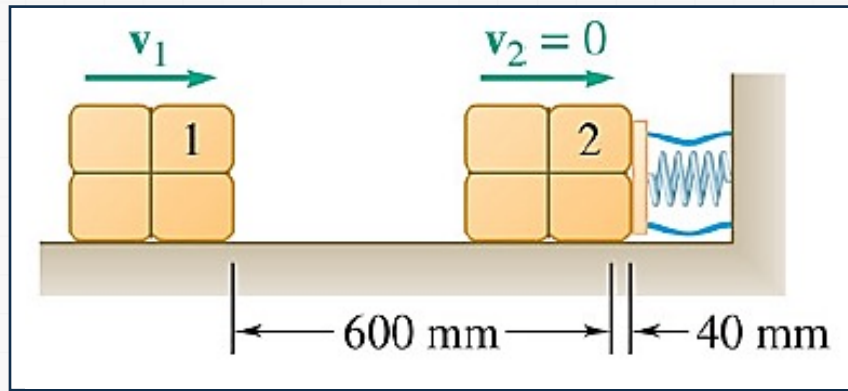
- Apply the principle of work and energy between the initial position and the point at which the spring is fully compressed and the velocity is zero. The only unknown in the relation is the friction coefficient.
- Apply the principle of work and energy for the rebound of the package. The only unknown in the relation is the velocity at the final position.

6 Work and Energy Principle

Sample Problem 2

⚙ Modeling and Analysis

- Apply principle of work and energy between initial position and the point at which spring is fully compressed.



$$T_1 = \frac{1}{2}mv_1^2 = \frac{1}{2}(60)(2.5)^2 = 187.5J$$

$$T_2 = 0$$

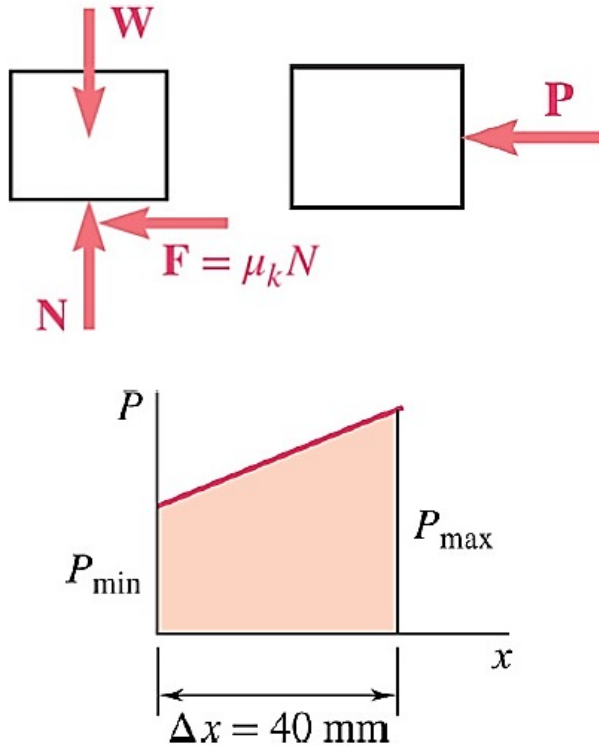
$$(U_{1 \rightarrow 2})_f = -\mu_k W_x$$

$$= \mu_k(60)(9.81)(0.640) = -(377)\mu_k$$

6 Work and Energy Principle

Sample Problem 2

Modeling and Analysis



$$P_{min} = kx_0 = (20,000)(0.120) = 2400N$$

$$P_{max} = k(x_0 + \Delta x) = (20,000)(0.120) = 2400N$$

$$\begin{aligned} (U_{1 \rightarrow 2})_e &= -\frac{1}{2}(P_{min} + P_{max})\Delta x \\ &= -\frac{1}{2}(2400 + 3200)(0.040) = -112.0J \end{aligned}$$

$$U_{1 \rightarrow 2} = (U_{1 \rightarrow 2})_f + (U_{1 \rightarrow 2})_e = -(377)\mu_k - 112J$$

$$T_1 + U_{1 \rightarrow 2} = T_2:$$

$$187.5 - (377)\mu_k - 112 = 0$$

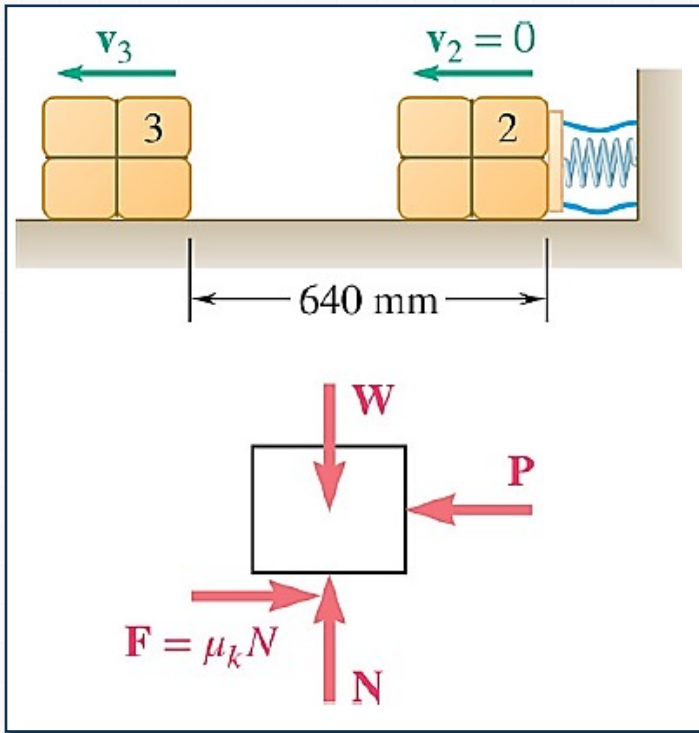
$$\mu_k = 0.20$$

6 Work and Energy Principle

Sample Problem 2

Modeling and Analysis

- Apply the principle of work and energy for the rebound of the package.



$$T_2 = 0 \qquad T_3 = \frac{1}{2} m v_3^2 = \frac{1}{2} (60 \text{ kg}) v_3^2$$

$$U_{2 \rightarrow 3} = (U_{2 \rightarrow 3})_f + (U_{2 \rightarrow 3})_e = -(377) \mu_k + 112$$

$$= +36.5 \text{ J}$$

$$T_2 + U_{2 \rightarrow 3} = T_3:$$

$$0 + 36.5 = \frac{1}{2} (60) v_3^2$$

$$v_3 = 1.103 \text{ m/s}$$

6 Work and Energy Principle

Sample Problem 2

⚙️ Reflect and Think

You needed to break this problem into two segments. From the first segment you were able to determine the coefficient of friction. Then you could use the principle of work and energy to determine the velocity of the package at any other location.

6 Work and Energy Principle

Sample Problem 2

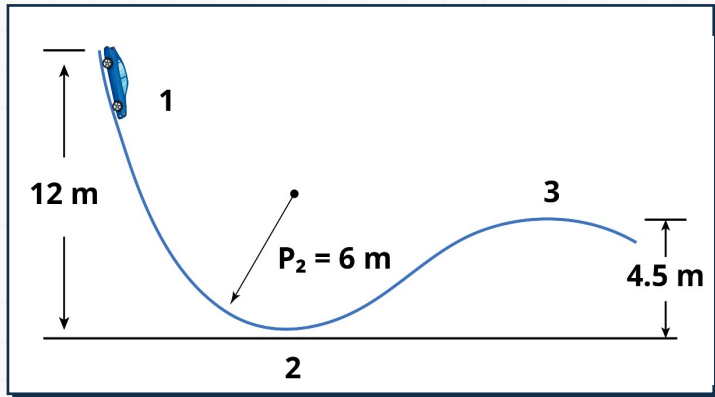
⚙️ Reflect and Think

Note that the system does not lose any energy due to the spring; it returns all of its energy back to the package. You would need to design something that could absorb the kinetic energy of the package in order to bring it to rest.

6 Work and Energy Principle

Sample Problem 3

⚙️ A 1000 kg car starts from rest at point 1 and moves without friction down the track shown.



• Determine

- a) the force exerted by the track on the car at point 2
- b) the minimum safe value of the radius of curvature at point 3.

6 Work and Energy Principle

Sample Problem 3

⚙️ **A 1000 kg car starts from rest at point 1 and moves without friction down the track shown.**

• **Strategy**

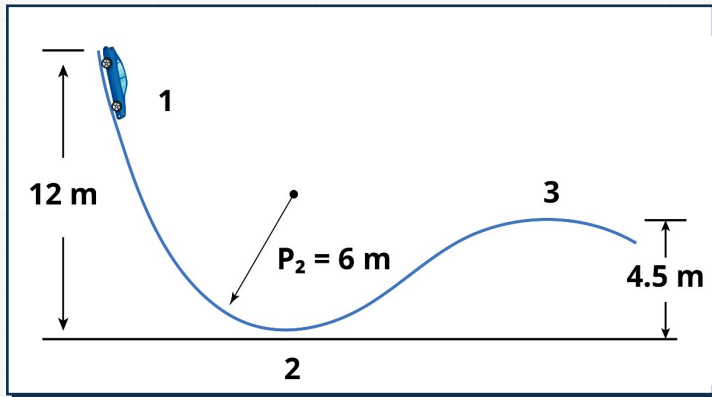
- Apply principle of work and energy to determine velocity at point 2.
- Apply Newton's second law to find normal force by the track at point 2.
- Apply principle of work and energy to determine velocity at point 3.
- Apply Newton's second law to find minimum radius of curvature at point 3 such that a positive normal force is exerted by the track.

6 Work and Energy Principle

Sample Problem 3

⚙ Modeling and Analysis

- Apply principle of work and energy to determine velocity at point 2.



$$T_1 = 0 \quad T_2 = \frac{1}{2}mv_2^2$$

$$U_{1 \rightarrow 2} = +mg(12)$$

$$T_1 + U_{1 \rightarrow 2} = T_2: \quad 0 + mg(12) = \frac{1}{2}mv_2^2$$

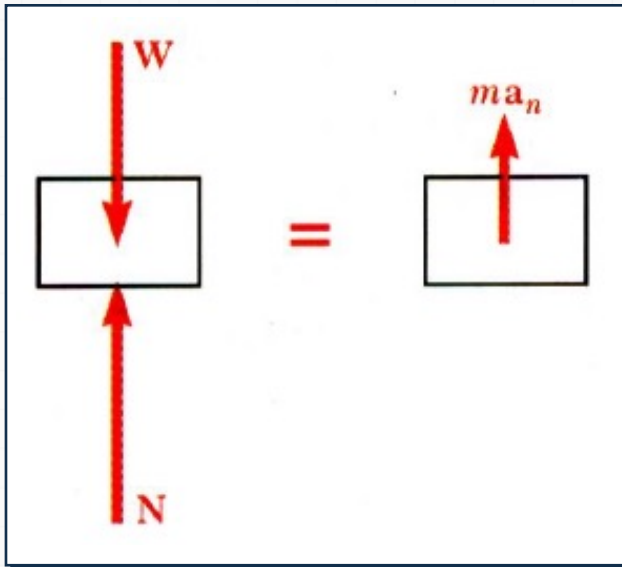
$$v_2^2 = 2(12)g = 2(12)(9.81) \quad v_2 = 15.34 \text{ m/s}$$

6 Work and Energy Principle

Sample Problem 3

⚙ Modeling and Analysis

- Apply Newton's second law to find normal force by the track at point 2.



$$+\uparrow \sum F_N = ma_n:$$

$$-W + N = ma_n = m \frac{v_2^2}{\rho_2} = m \frac{(24g)}{6} = 4mg = 4W$$

$$N = 5W(1000)(9.81)$$

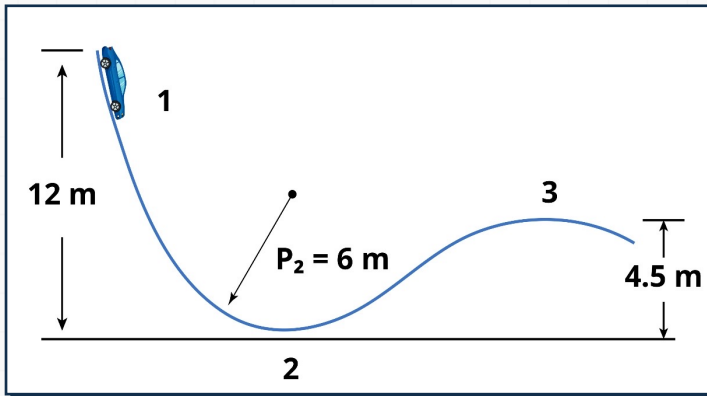
$$N = 49.05kN$$

6 Work and Energy Principle

Sample Problem 3

⚙ Modeling and Analysis

- Apply principle of work and energy to determine velocity at point 3.



$$T_1 + U_{1 \rightarrow 3} = T_3 \quad 0 + mg(7.5) = \frac{1}{2}mv_3^2$$

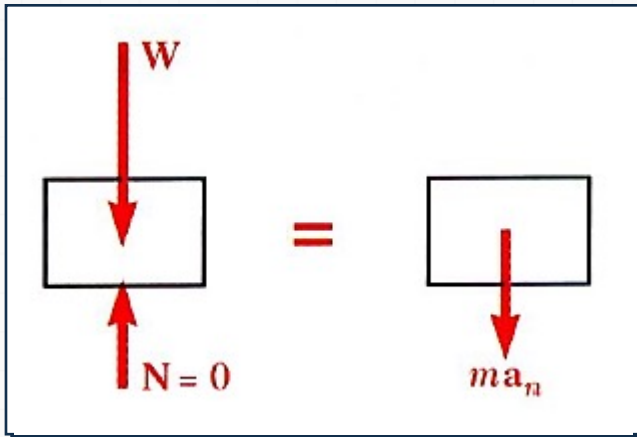
$$v_3^2 = 2(7.5)g = 2(7.5)(9.81) \quad v_3 = 12.13 \text{ m/s}$$

6 Work and Energy Principle

Sample Problem 3

⚙ Modeling and Analysis

- Apply principle of work and energy to determine velocity at point 3.



$$+\downarrow \sum F_n = ma_n:$$

$$W = ma_n$$

$$= m \frac{v_3^2}{\rho_3} = m \frac{2(7.5)g}{\rho_3}$$

$$\rho_3 = 15m$$

6 Work and Energy Principle

Sample Problem 3

⚙️ Reflect and Think

This is an example where you need both Newton's second law and the principle of work and energy.

Work-energy is used to determine the speed of the car, and Newton's second law is used to determine the normal force.

6 Work and Energy Principle

Sample Problem 3

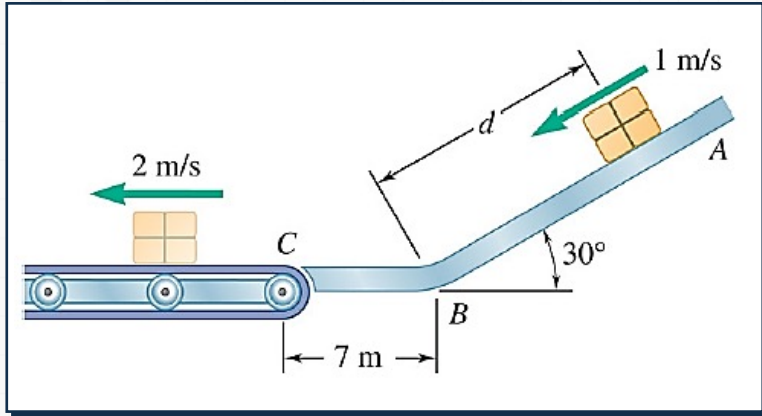
⚙️ Reflect and Think

A normal force of $5W$ is equivalent to a fighter pilot pulling $5g$'s and should only be experienced for a very short time.

For safety, you would also want to make sure your radius of curvature was quite a bit larger than $15m$.

6 Work and Energy Principle

Group Problem Solving



Packages are thrown down an incline at A with a velocity of 1 m/s . The packages slide along the surface ABC to a conveyor belt which moves with a velocity of 2 m/s . Knowing that $\mu_k = 0.25$ between the packages and the surface ABC , determine the distance d if the packages are to arrive at C with a velocity of 2 m/s .

6 Work and Energy Principle

Group Problem Solving

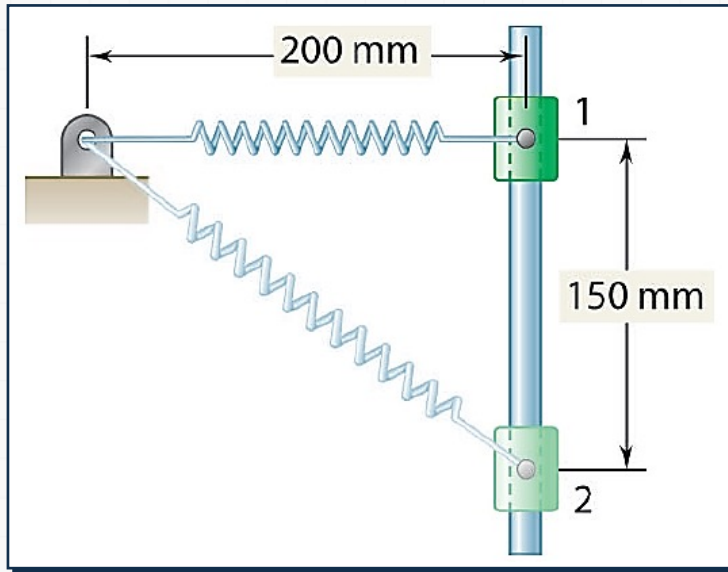
⚙ Strategy

- The problem deals with a change in position and different velocities, so use work-energy.
 - Draw *FBD* of the box to help us determine the forces that do work.
 - Determine the work done between points *A* and *C* as a function of *d*.
 - Find the kinetic energy at points *A* and *C*.
 - Use the work-energy relationship and solve for *d*.

6 Work and Energy Principle

Sample Problem 4

⚙️ A 10kg collar slides without friction along a vertical rod as shown.



• Determine

- The spring attached to the collar has an undeflected length of 100 mm and a constant of 200 N/m .
- If the collar is released from rest at position 1, determine its velocity after it has moved 150 mm to position 2.

6 Work and Energy Principle

Sample Problem 4

⚙️ **A 10kg collar slides without friction along a vertical rod as shown.**

• **Strategy**

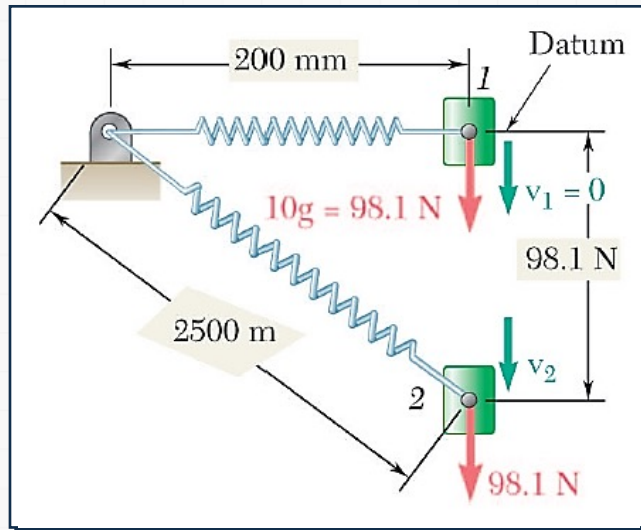
- Apply the principle of conservation of energy between positions 1 and 2.
- The elastic and gravitational potential energies at 1 and 2 are evaluated from the given information. The initial kinetic energy is zero.
- Solve for the kinetic energy and velocity at 2.

6 Work and Energy Principle

Sample Problem 4

⚙ Modeling and Analysis

- Apply the principle of conservation of energy between positions 1 and 2.



■ Position 1

$$V_e = \frac{1}{2} k x_1^2 = \frac{1}{2} (200)(0.2 - 0.1)^2 = 1 \text{ N} \cdot \text{m}$$

$$V_1 = V_e + V_g = 1 + 0 = 1 \text{ N} \cdot \text{m}$$

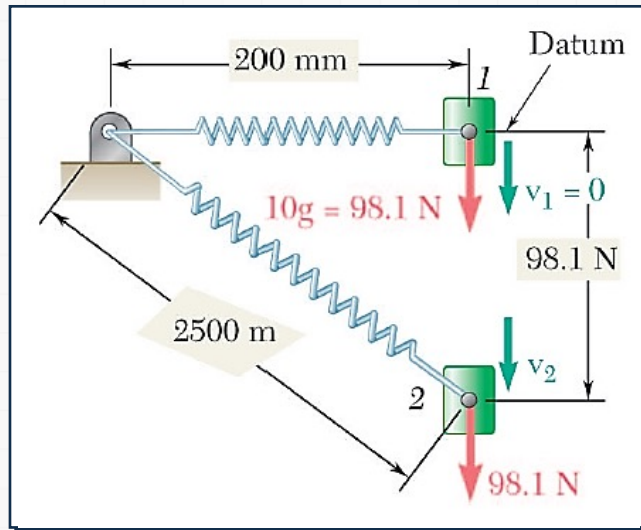
$$T_1 = 0$$

6 Work and Energy Principle

Sample Problem 4

⚙ Modeling and Analysis

- Apply the principle of conservation of energy between positions 1 and 2.



■ Position 2

$$V_e = \frac{1}{2} k x_2^2 = \frac{1}{2} (200) (0.25 - 0.1)^2 = 2.25 \text{ N} \cdot \text{m}$$

$$V_g = W_y = (9.81) (-0.15) = -14.715 \text{ N} \cdot \text{m}$$

$$V_2 = V_e + V_g = 2.25 - 14.715 = -12.465 \text{ N} \cdot \text{m}$$

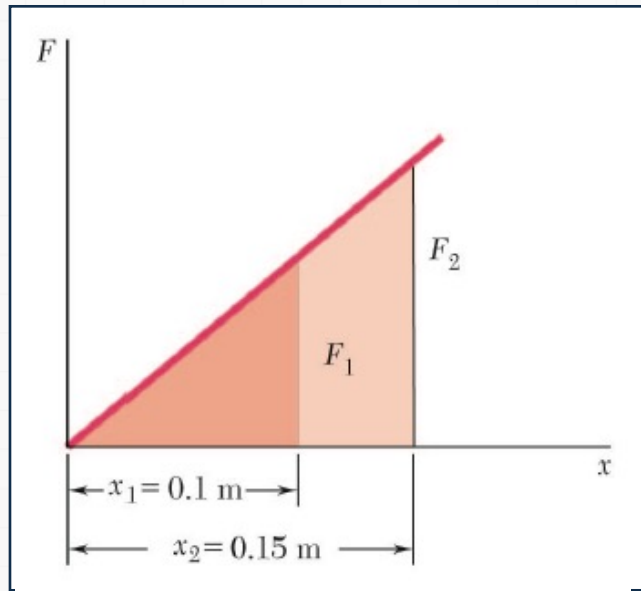
$$T_2 = \frac{1}{2} m v_2^2 = \frac{1}{2} (10) v_2^2 = 5 v_2^2$$

6 Work and Energy Principle

Sample Problem 4

⚙ Modeling and Analysis

- Apply the principle of conservation of energy between positions 1 and 2.



- Conservation of Energy

$$T_1 + V_1 = T_2 + V_2$$

$$0 + 1 \text{ N} \cdot \text{m} = 5v_2^2 - 12.465 \text{ N} \cdot \text{m}$$

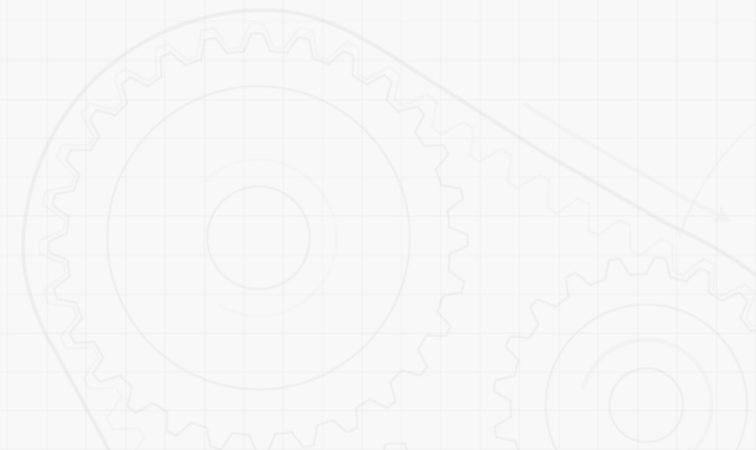
$$v_2 = 1.641 \text{ m/s} \downarrow$$

6 Work and Energy Principle

Sample Problem 4

⚙️ Reflect and Think

If you had not included the spring in your system, you would have needed to treat it as an external force; therefore, you would have needed to determine the work.



6 Work and Energy Principle

Sample Problem 4

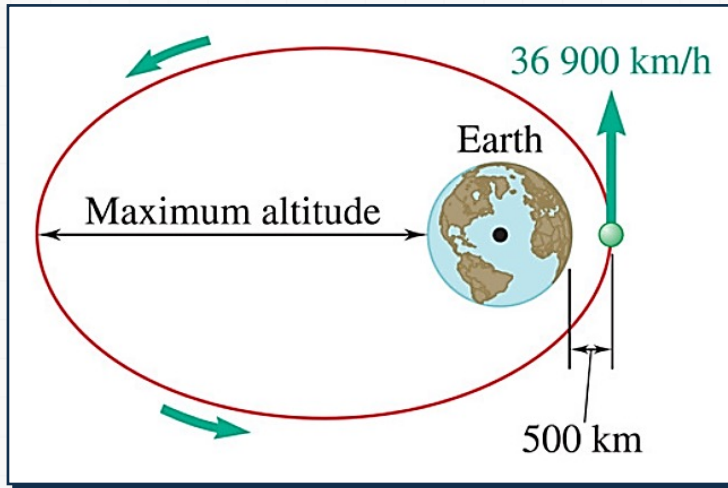
⚙️ Reflect and Think

Similarly, if there was friction acting on the collar, you would have needed to use the more general work-energy principle to solve this problem. It turns out that the work done by friction is not very easy to calculate because the normal force depends on the spring force.

6 Work and Energy Principle

Sample Problem 5

⚙️ A satellite is launched in a direction parallel to the surface of the earth with a velocity of 36900 km/h from an altitude of 500 km .



• Determine

- Determine (a) the maximum altitude reached by the satellite.
- Determine (b) the maximum allowable error in the direction of launching if the satellite is to come no closer than 200 km to the surface of the earth.

6 Work and Energy Principle

Sample Problem 5

⚙️ **A satellite is launched in a direction parallel to the surface of the earth with a velocity of 36900 km/h from an altitude of 500km.**

• **Strategy**

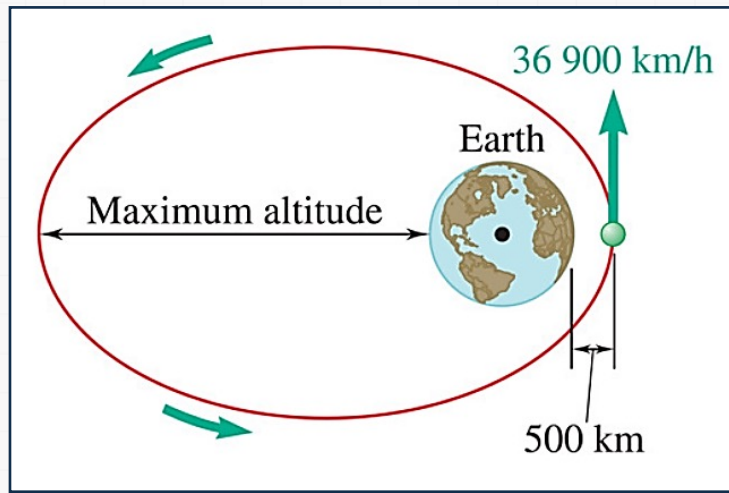
- For motion under a conservative central force, the principles of conservation of energy and conservation of angular momentum may be applied simultaneously.
- Apply the principles to the points of minimum and maximum altitude to determine the maximum altitude.
- Apply the principles to the orbit insertion point and the point of minimum altitude to determine maximum allowable orbit insertion angle error.

6 Work and Energy Principle

Sample Problem 5

⚙️ Modeling and Analysis

- Apply the principles of conservation of energy and conservation of angular momentum to the points of minimum and maximum altitude to determine the maximum altitude.



- Conservation of energy

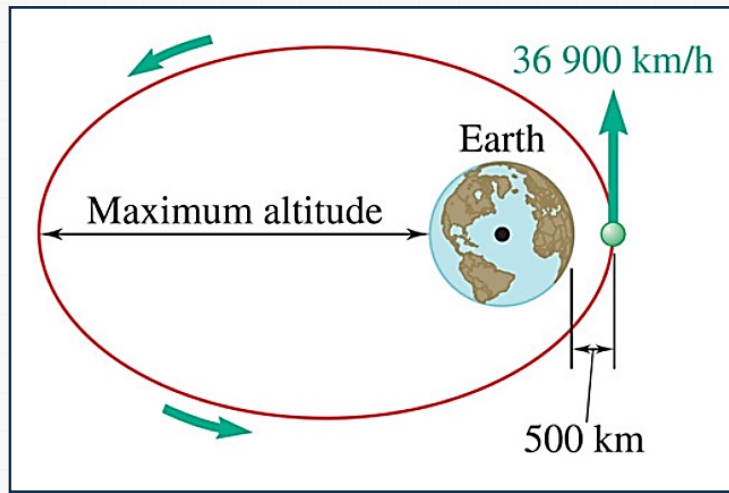
$$T_A + V_A = T_{A'} + V_{A'} \quad \frac{1}{2}mv_0^2 - \frac{GMm}{r_0} = \frac{1}{2}mv_1^2 - \frac{GMm}{r_1}$$

6 Work and Energy Principle

Sample Problem 5

⚙ Modeling and Analysis

- Apply the principles of conservation of energy and conservation of angular momentum to the points of minimum and maximum altitude to determine the maximum altitude.



- Conservation of angular momentum

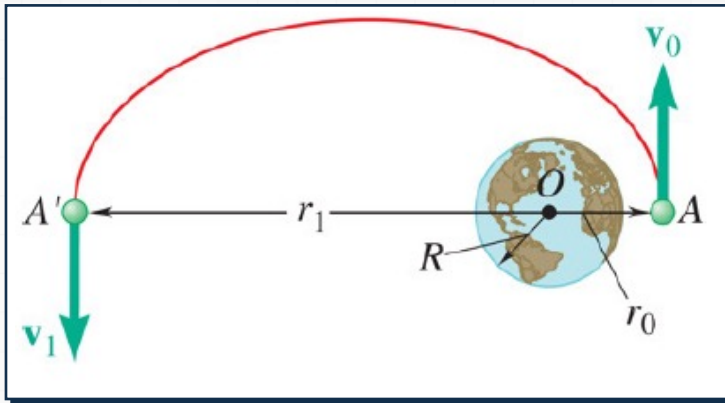
$$r_0 m v_0 = r_1 m v_1 \qquad v_1 = v_0 \frac{r_0}{r_1}$$

6 Work and Energy Principle

Sample Problem 5

Modeling and Analysis

- Apply the principles of conservation of energy and conservation of angular momentum to the points of minimum and maximum altitude to determine the maximum altitude.



Combining

$$\frac{1}{2} v_0^2 \left(1 - \frac{r_0^2}{r_1^2}\right) = \frac{GM}{r_0} \left(1 - \frac{r_0}{r_1}\right) \quad 1 + \frac{r_0}{r_1} = \frac{2GM}{r_0 v_0^2}$$

$$r_0 = 6370 + 500 = 6870 \text{ km}$$

$$v_0 = 36900 = 10.25 \times 10^3 \text{ m/s}$$

$$GM = gR^2 = (9.81)(6.37 \times 10^6)^2 = 398 \times 10^{12} \text{ m}^3/\text{s}^2$$

$$r_1 = 66.8 \times 10^6 \text{ m}$$

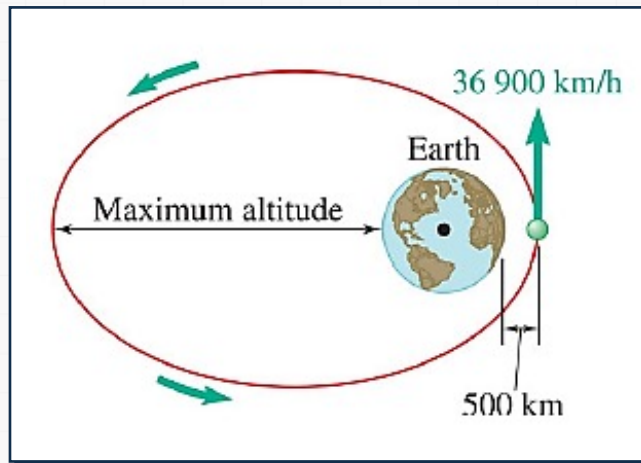
$$\text{Max alt} = r_1 - 6.37 \times 10^6 = 60.4 \times 10^6$$

6 Work and Energy Principle

Sample Problem 5

⚙ Modeling and Analysis

- Apply the principles to the orbit insertion point and the point of minimum altitude to determine maximum allowable orbit insertion angle error.



- Conservation of energy

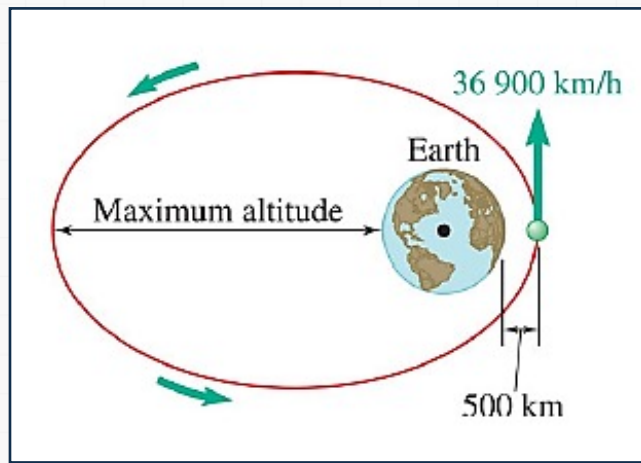
$$T_0 + V_0 = T_A + V_A \quad \frac{1}{2}mv_0^2 - \frac{GMm}{r_0} = \frac{1}{2}mv_{\max}^2 - \frac{GMm}{r_{\min}}$$

6 Work and Energy Principle

Sample Problem 5

Modeling and Analysis

- Apply the principles to the orbit insertion point and the point of minimum altitude to determine maximum allowable orbit insertion angle error.



- Conservation of angular momentum

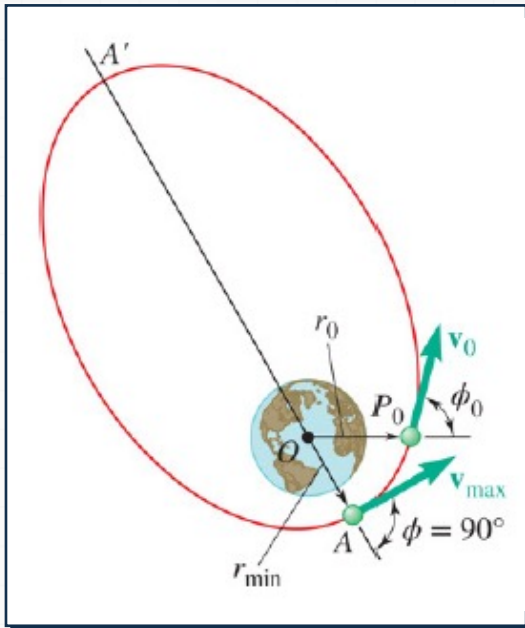
$$r_0 m v_0 \sin \phi_0 = r_{min} m v_{max} \quad v_{max} = v_0 \sin \phi_0 \frac{r_0}{r_{min}}$$

6 Work and Energy Principle

Sample Problem 5

⚙ Modeling and Analysis

- Apply the principles to the orbit insertion point and the point of minimum altitude to determine maximum allowable orbit insertion angle error.



- Combining and solving for $\sin \phi_0$,

$$\sin \phi_0 = 0.9801$$

$$\phi_0 = 90^\circ \pm 11.5^\circ$$

$$\text{allowable error} = \pm 11.5^\circ$$

6 Work and Energy Principle

Sample Problem 5

⚙️ Reflect and Think

Space probes and other long-distance vehicles are designed with small rockets to allow for mid-course corrections.

Satellites launched from the Space Station usually do not need this kind of fine-tuning.

6 Work and Energy Principle

References

- Beer, Johnston, Cornwell, Self, Sanghi, "Vector Mechanics for Engineers: Dynamics" 12th Edition, Chapter 13